



THE UNIVERSITY OF
MELBOURNE

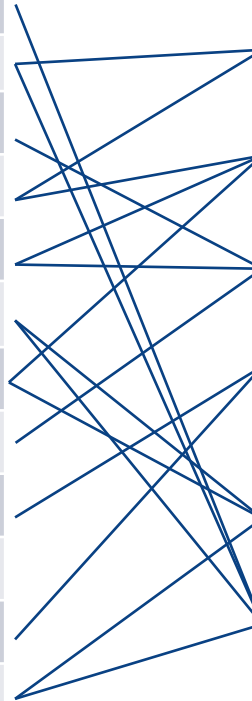
Probability Reliability and Quality

Hypothesis Testing Sample Sizes

Kevin Otto

Schedule and Learning Outcomes

Week	Recorded Lecture Topic
1	Introduction.
2	Inspection. Yield.
3	ANOVA. Measurement Systems Analysis.
4	Statistics. Confidence Intervals.
5	Hypothesis Testing. Sample Sizes.
6	Quality Management Processes.
7	Regression.
8	Design of Experiments.
9	System Reliability. Availability.
	<i>BREAK – Non Teaching Week</i>
10	Parametric Distribution Analysis
11	Statistical Process Control. Supplier Quality.
12	Review



After completion of the course the student should be able to:

- Explain different statistical methods
- **Parametric and non-parametric hypothesis tests**
- **Design and plan for collecting experimental data**
- Concepts of reliability engineering and the relationship between TTF, R, H
- How to improve its processes, products and services
- Apply TQM, 6S, and benchmarking



Overview

Test Errors

Statistical Inferencing

T-test

Parametric tests

Sample Sizes

Non-parametric tests

Methodology Road Map for Tool Use

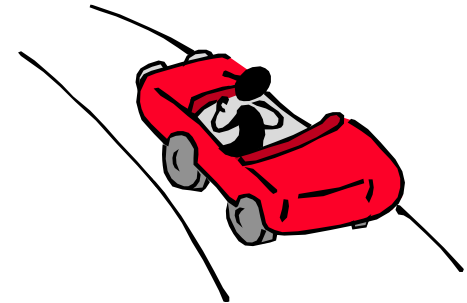
General use for Hypothesis Tests:

Use when...

- you need to compare the performance level of a mean, variance, or proportion against a stated (target) value.
- you need to compare the difference in the performance level of a mean, variance, or proportion between two sets.

Examples:

- Root cause analysis of defects
- Lot inspection
- Supplier certification
- Supplier selection
- ...





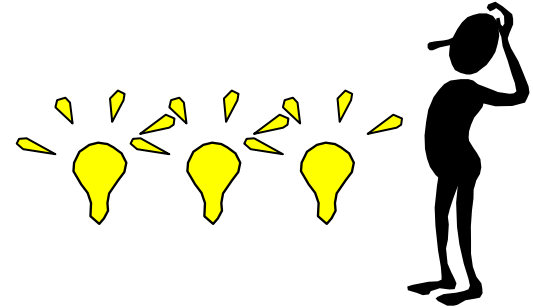
Hypothesis Testing Example

Single Mean Compared to Target

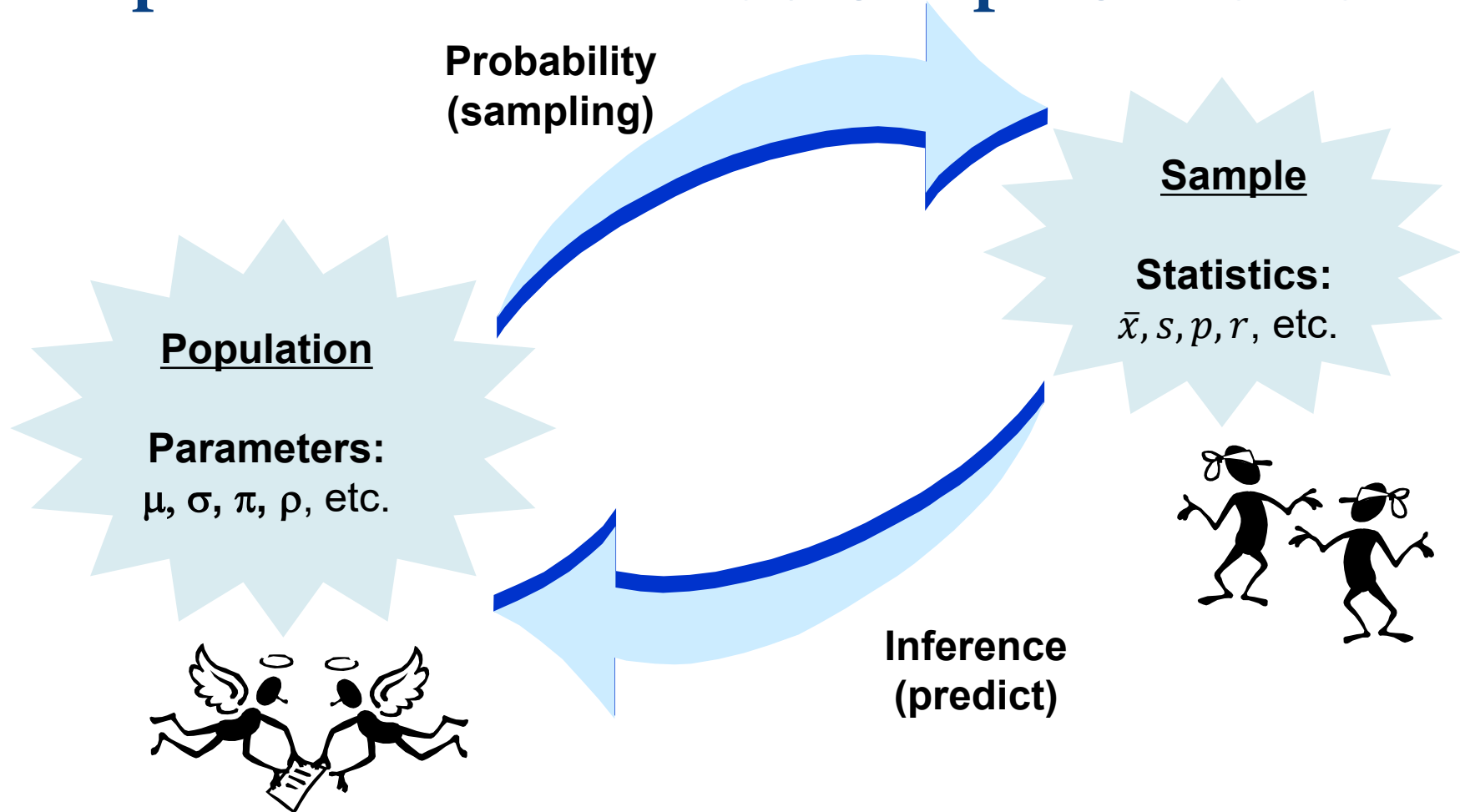
A supplier gives an average efficiency rating for a particular lamp at 86.5 percent. A random sample of 10 lamps provides the efficiency levels measured below.

86.61	86.24	86.41	86.70	86.50
86.38	86.45	86.52	86.28	86.75

One question is: Is the mean of the sample representative of a target value of 86.5%?



Population Parameters & Sample Statistics



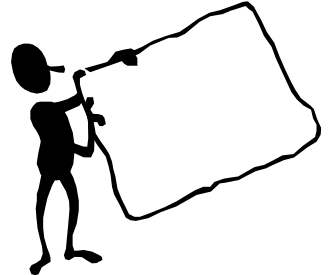


What is a Hypothesis?

A **hypothesis** is essentially just a claim. It is a statement about a parameter (μ, σ, π , etc.) of one or more populations.

Example Hypothesis: Average bulb efficiency is 86.5 percent

The purpose of the hypothesis is to establish a true/false statement, so that we can gather evidence as a random experiment, to either reject the claim or accept it as true



Hypotheses: Manufacturing Examples

In plain language...

This process change will reduce particle count defect

Increasing this variable will reduce performance variation

Eliminating this variation will increase factory yield 2%

This change will improve IT response time by 5 minutes

This process change will reduce billing errors

...



The Statistical Null Hypothesis

The statistical **null hypothesis** is often logically different than the practical hypothesis

A Null Hypothesis must be stated as a “no difference” claim

H_0 is called the Null Hypothesis

H_A is called the ***Alternative Hypothesis***

“There is no difference between the population mean and 86.50”

$$H_0: \mu = 86.50$$

$$H_{a1}: \mu > 86.50$$

$$H_{a2}: \mu < 86.50$$

$$H_{a3}: \mu \neq 86.50$$

Null hypotheses are always expressed mathematically with an “=”, never with a “>, ≥, <, ≤ or ≠”.



Hypothesis Testing Example

“Is the average efficiency $<$ my requirement of 86.50?”

The hypothesis can be stated succinctly as...

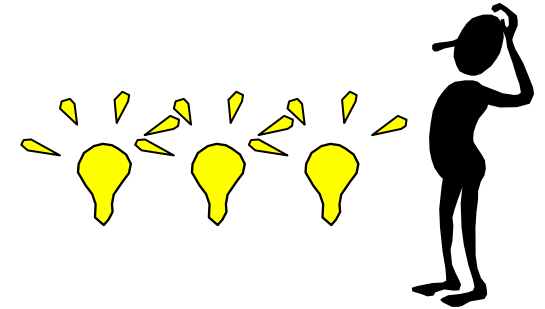
$$H_0: \mu \text{ efficiency} = 86.50$$

$$H_A: \mu \text{ efficiency} < 86.50$$

where:

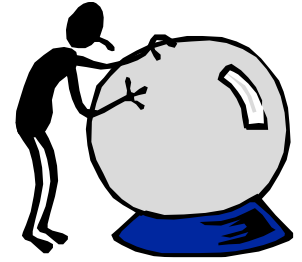
H_0 is called the ***Null Hypothesis***

H_A is called the ***Alternative Hypothesis***



Note that H_0 is not always what you are interested in. You may be more interested in H_A .

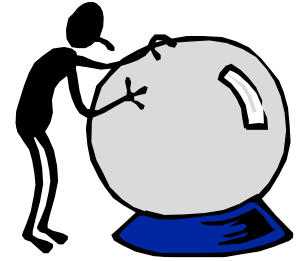
Test vs. Reality



There are four possible outcomes associated with our hypothesis...

		Reality	
		H_0 TRUE	H_0 FALSE
Decision	ACCEPT H_0	Correct Acceptance	Type II Error
	REJECT H_0	Type I Error	Correct Rejection

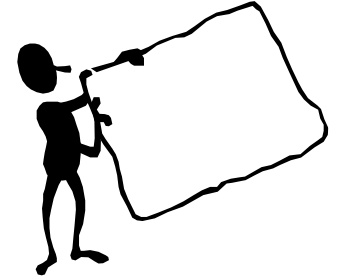
Test vs. Reality



There are four possible outcomes associated with our hypothesis...

		Reality	
		H_0 TRUE	H_0 FALSE
Decision	ACCEPT H_0	Correct Acceptance CONFIDENCE $1 - \alpha$	Type II Error Beta Risk β
	REJECT H_0	Type I Error Alpha Risk α	Correct Rejection POWER $1 - \beta$

Test vs. Reality



Confidence:

- How sure you are of accepting H_0 , GIVEN THAT IT IS TRUE.

Alpha risk:

- Your risk in rejecting H_0 , GIVEN THAT IT IS TRUE.
- Sometimes called ***Producers Risk***.

Power:

- How sure you are of rejecting H_0 , GIVEN THAT IT IS FALSE.

Beta risk:

- Your risk in accepting H_0 , GIVEN THAT IT IS FALSE.
- Sometimes called ***Consumer's Risk***.

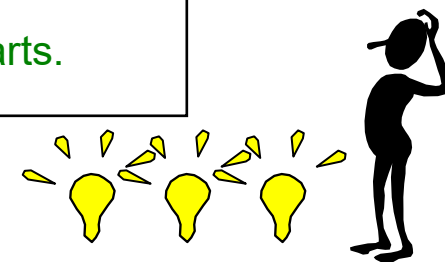
Hypothesis Testing Example

Consider the lamp example. What value of α and β should you use to accept?

Reality: Are the supplied lamps good?

	H_0 TRUE	H_0 FALSE
ACCEPT H_0	Correct Acceptance CONFIDENCE $1 - \alpha$ Accept good parts.	Type II Error Beta Risk β Bad parts go into products.
REJECT H_0	Type I Error Alpha Risk α Scrap good parts. Costs money...	Correct Rejection POWER $1 - \beta$ Reject bad parts.

Test Result:
Lamps are good.



Test vs. Reality: Pacemaker Acceptance

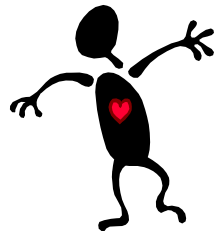
Suppose you manufacture Pace Makers. What value of α and β should you use at final unit test?

Reality: Is the pacemaker is good?

ACCEPT H_0

Test Result:
Pacemaker is good.

REJECT H_0



	H_0 TRUE	H_0 FALSE
ACCEPT H_0	Correct Acceptance CONFIDENCE $1 - \alpha$ Accept good part.	Type II Error Beta Risk β Bad part goes into a patient.
REJECT H_0	Type I Error Alpha Risk α Scrap good part. Costs money...	Correct Rejection POWER $1 - \beta$ Reject bad part.

Typical Values for α and β

How likely is the error to occur?

Alpha / Beta		Occurrence				
		1 – 2	3 – 4	5 – 6	7 – 8	9 – 10
Severity	1 – 2	10 / 20	10 / 15	10 / 10	5 / 10	5 / 10
	3 – 4	10 / 15	10 / 10	5 / 10	5 / 10	5 / 5
	5 – 6	10 / 10	5 / 10	5 / 10	5 / 5	3 / 5
	7 – 8	5 / 10	5 / 10	5 / 5	5 / 3	1 / 5
	9 – 10	5 / 10	5 / 5	5 / 3	1 / 3	1 / 1

If the error occurs,
what is the impact?



Hypothesis Testing – General 10 Step Procedure

1. State the Practical Problem
2. State the test and it's Null and Alternative Hypotheses
3. Define risk levels using alpha α & beta β
4. Determine sample size
5. Randomly sample from the population & Measure
6. Check to ensure the assumptions of the test
 - For example, normality, equal variances, ...
7. Calculate the model result and it's test-statistic value
8. Assess adequacy of the model result
9. Determine the test statistic's p-value and make a statistically correct decision
 - If $p\text{-value} < \alpha$, then REJECT H_0
10. Translate the statistical conclusion into practical terms



What Is a P-Value?

p – value:

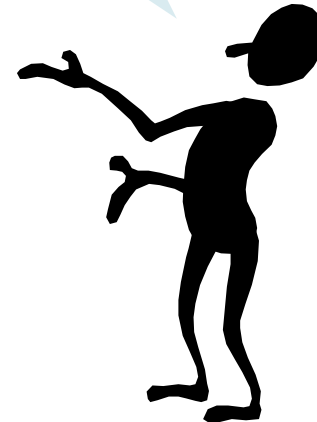
Rigorous definition: The likelihood that the test statistic value observed occurs by chance.

Practical definition: The chance of being wrong when you reject the null hypothesis.

The Rule:

If p is less than α ,
then reject the null hypothesis!

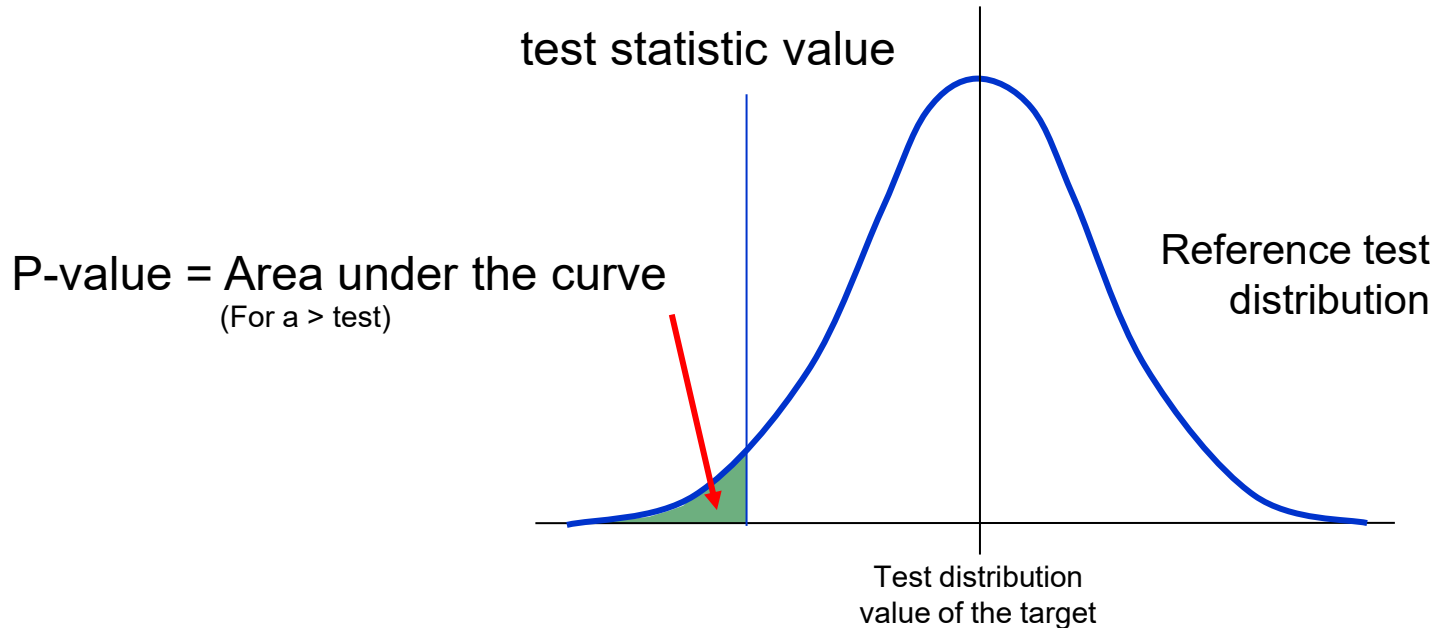
“If p is low, H-oh must go”



What Is a P-Value (Restated)

The probability you've made a Type I error (incorrectly rejecting H_0).

This is the probability of getting a value as large as or larger than the statistic observed, given that H_0 is true.



Hypothesis Testing Example: Steps 1-3

Step 1: State the Practical Problem

- Should we reject the supplier: Is the population mean less than our target value of 86.5%?

Step 2: State the statistical test and hypotheses

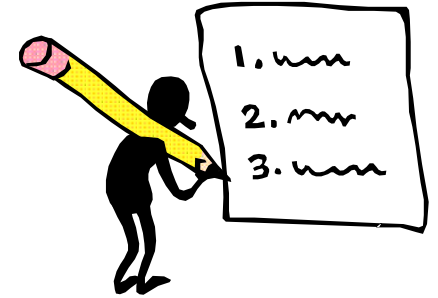
- t-test for 1 population, using the t-statistic
- $H_0: \mu_{\text{efficiency}} = 86.5$
- $H_A: \mu_{\text{efficiency}} < 86.5$

Step 3: State maximum acceptable alpha and beta risks

- $\alpha = 0.05$
- $\beta = 0.05$



t-test



The t-test is used in the following applications:

- Single mean compared to a target value
- Comparison of two independent group means
- Paired Comparison of two dependent group means

...when the standard deviation is unknown!

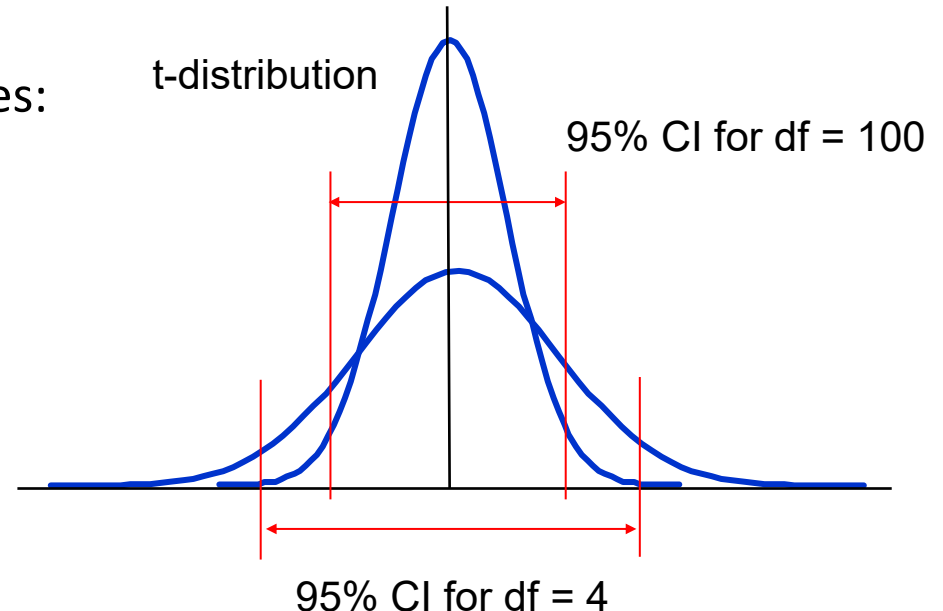
Review: What was the t-Distribution?

The **t-Distribution** is a family of normal-shaped distributions that are dependent on sample size

- The shape depends on the sample size n . The smaller n , the “flatter” is the t distribution, compared to Z
- Degrees of freedom = $df = v = n - 1$ the number from the sample that are free to vary after the sample statistic has been calculated

Values of t for 95% CI's for different sample sizes:

Sample	t-value
5	2.78
10	2.26
20	2.09
30	2.05
100	1.98
1000	1.96



Review: What was the t-statistic?

The ***t-statistic*** is the statistic associated with a t-distribution

- Similar to a Z-score for the normal distribution
- The t-statistic is defined as

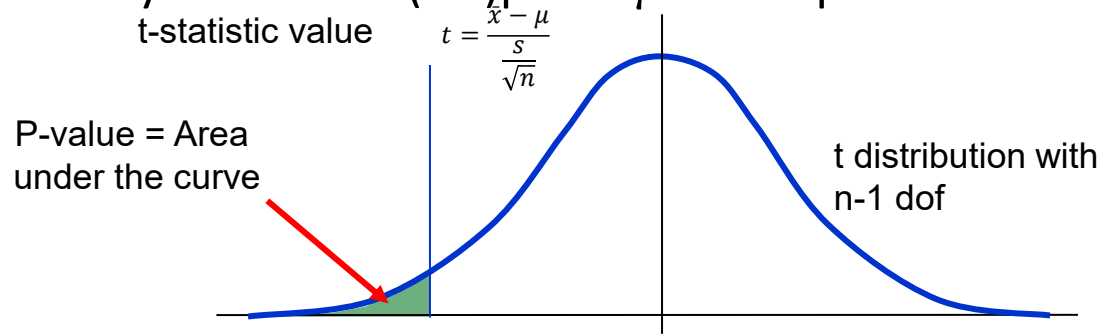
$$t = \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}}$$

FYI: all test statistics are very similar

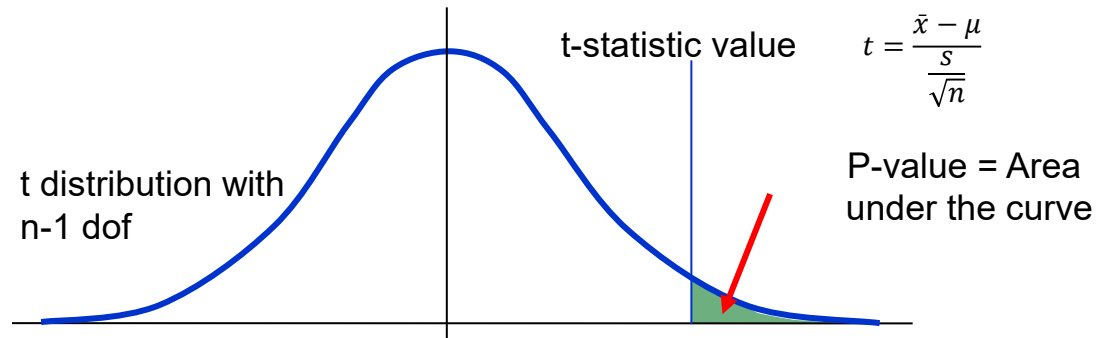
- They always have an interesting gap in the numerator
- They always have a measure of noise in the denominator
- A test statistic is always a “signal to noise” measure

P-value of a 1-sided t-test

$H_A: \mu > T$: Use when you want to (dis)prove $\mu > \text{a requirement } T$

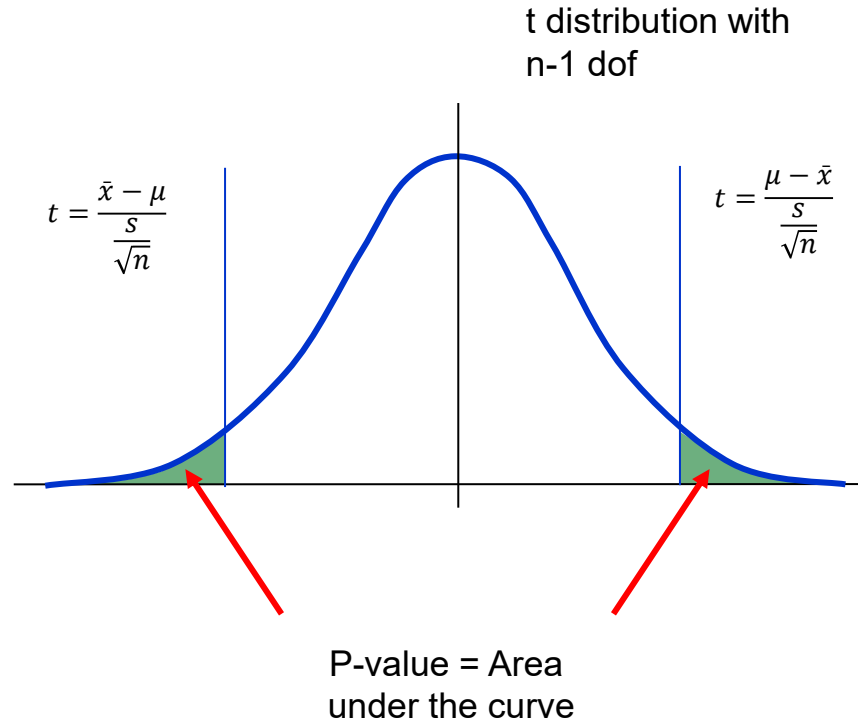


$H_A: \mu < T$: Use when you want to (dis)prove $\mu < \text{a requirement } T$



P-value of a 2-sided t-test

$H_A: \mu \neq T$: Use when you want to (dis)prove $\mu \neq$ a requirement T





One-Sided vs Two-Sided Tests

A test that is concerned only with whether the difference is greater than or less than (but not both) is called a ***one-sided*** test.

If the test involves inequality, then the test is called ***two sided***. In such cases, the α value is substituted by $\alpha/2$ in our calculations.

Hypothesis Testing Example: Steps 4 – 5

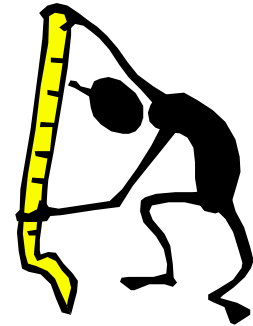
Step 4: Determine sample size

- We'll come back to this step later. For now use $n = 10$.

Step 5: Randomly sample from the population. Measure.

- Collect the samples
- Measure the data

86.4
86.3
86.8
86.4
86.6
86.7
86.0
86.6
86.2
86.3

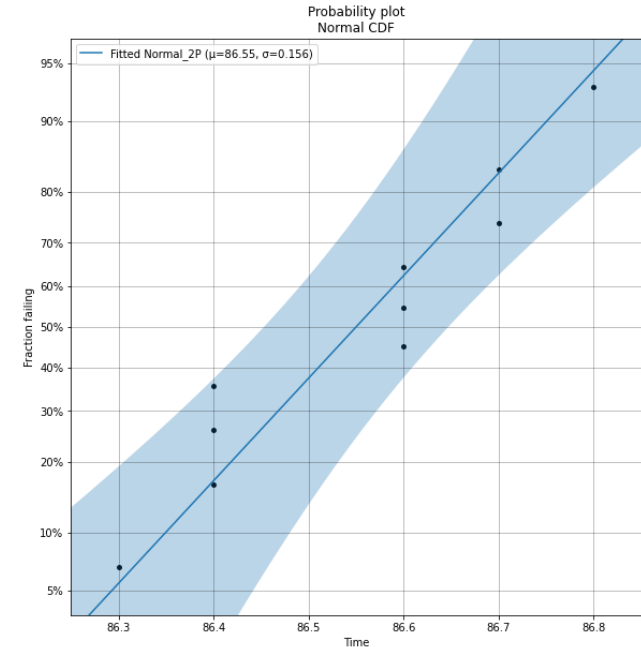


Hypothesis Testing Example: Step 6

Step 6: Check to ensure the data is normal

Minitab: **Stat > Basic Stat > Normality Test**

Python: `Normal_probability_plot(data)`



H_0 : The data is normally distributed.
 $p=0.810$ is not low. Do not reject the null.

Anderson Darling $AD2=0.209$
 $p\text{-value}=0.810$

Hypothesis Testing Example: Step 7 – 8

Step 7: Determine the sample's test statistic value

$$t = \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}}$$

$$t = \frac{86.43 - 86.5}{\frac{0.245}{\sqrt{10}}} = \frac{-0.07}{0.078} = -0.903$$

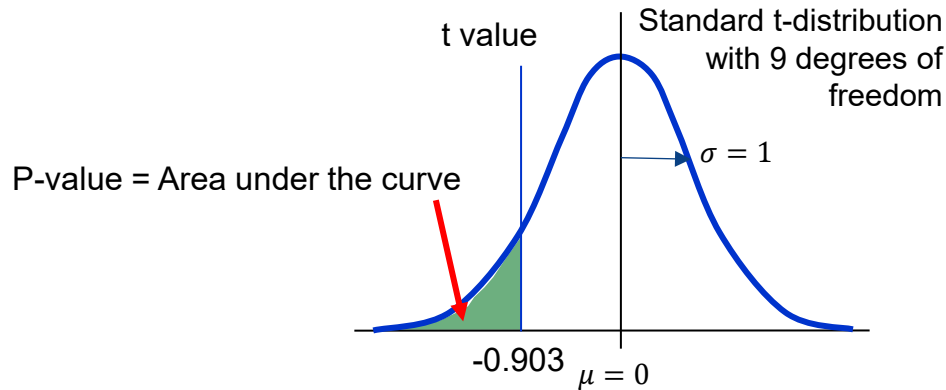
Hypothesis Testing Example: Step 9

Step 9: Find the test statistic's p-value

Here, we are computing for $H_A: \mu < T$, so the p-value is the cumulative integral of the standard t-distribution with 9 degrees of freedom for $t = -0.905$

Using Excel, Minitab or Python: `t.cdf(t, df, loc=0, scale=1)`

t stat = -0.903
p-value = 0.195



Hypothesis Testing Example: Step 9

Step 9: Make a statistical correct decision

If $p\text{-value} < \alpha$, then REJECT H_0

In our example,

If $0.195 < 0.05$ then reject H_0

p is not less than α , we cannot reject H_0



Hypothesis Testing Example: Step 10

Step 10: Translate the statistical conclusion into practical terms

- We fail to reject H_0
- We fail to reject " $\mu = 86.5\%$ "

We have sufficient statistical evidence to infer that the population mean from which the sample came is not significantly different than 86.5%.





Our Conclusion Statement

Since $p \geq \alpha$ (0.05 in this case), we can make the following statement:

- We have sufficient statistical evidence to infer that the population mean from which the sample came is not significantly different than 86.5.

Does this prove that the null hypothesis is true (that the true population mean = 86.5)?

- Unfortunately not...

We have evidence to “not reject” the null, but not to “prove” it.

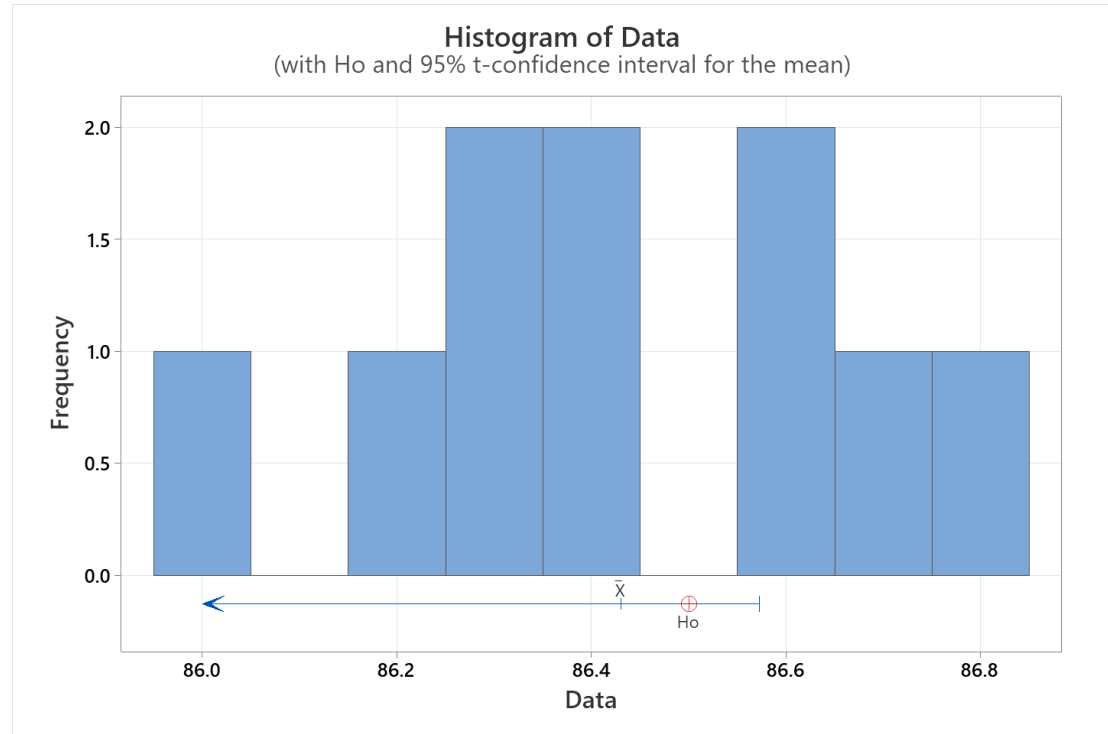
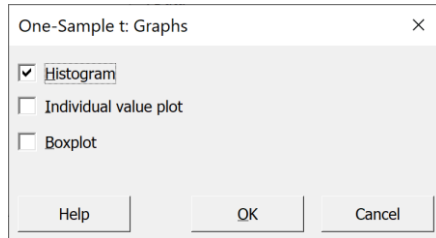
- Statistically, all we can ever do is not reject. We can never prove or accept.

However, we usually operate under the assumption that not rejecting is practically the same as accepting.

- So we “accept” that H_0 is true.

Graphical Interpretation: p-values and CIs

We can also graph the histogram of the data, show the null hypothesis target, and the one sided confidence interval. When the target is within the confidence interval, the p-value will be less than α and there is no difference.





Overview

Test Errors

Statistical Inferencing

T-test

Parametric tests

Sample Sizes

Non-parametric tests

Parametric Hypothesis Tests

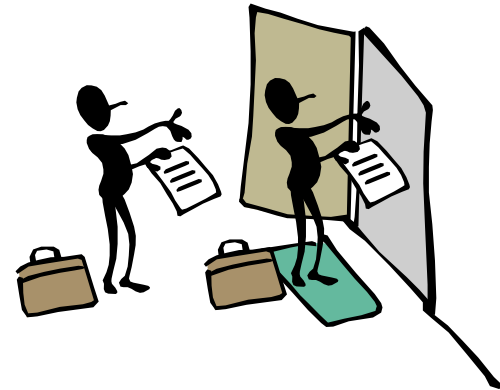
Test	Purpose	Null
Anderson-Darling test Kolmogorov-Smirnov test	Data follows a distribution	follows
One sample t-test	Mean vs a target	$\mu = \mu_0$
One sample χ^2 test	Variance vs a target	$\sigma^2 = \sigma_0^2$
Two sample F test Barlett's test Levene's test	Two variances	$\sigma_1^2 = \sigma_2^2$
Two sample t-test	Two means	$\mu_1 = \mu_2$
Paired t-test	Two means of paired data	$\mu_1 = \mu_2$
One sample proportion test	Proportion vs a target	$\pi = \pi_0$
Two sample proportion test	Two Proportions	$\pi_1 = \pi_2$
One sample rate test	Rate vs a target	$\rho = \rho_0$
Two sample rate test	Two rates	$\rho_1 = \rho_2$



Test a Single Variance vs. a Target Value

What about variation?

Lets look at an example.



Test a Single Variance vs. a Target Value

Problem:

The breaking strength of a plastic you use in your process is important. A new supplier, Manufacturer B, tells you that her plastic strength is 35 MPa $\pm$ 4 MPa. You ask what does that mean, and are informed batches are out of spec “about 1 in 100,000” or $\sim$ 4 sigma.

You would like to verify this claim.

Step 2: Hypothesis:

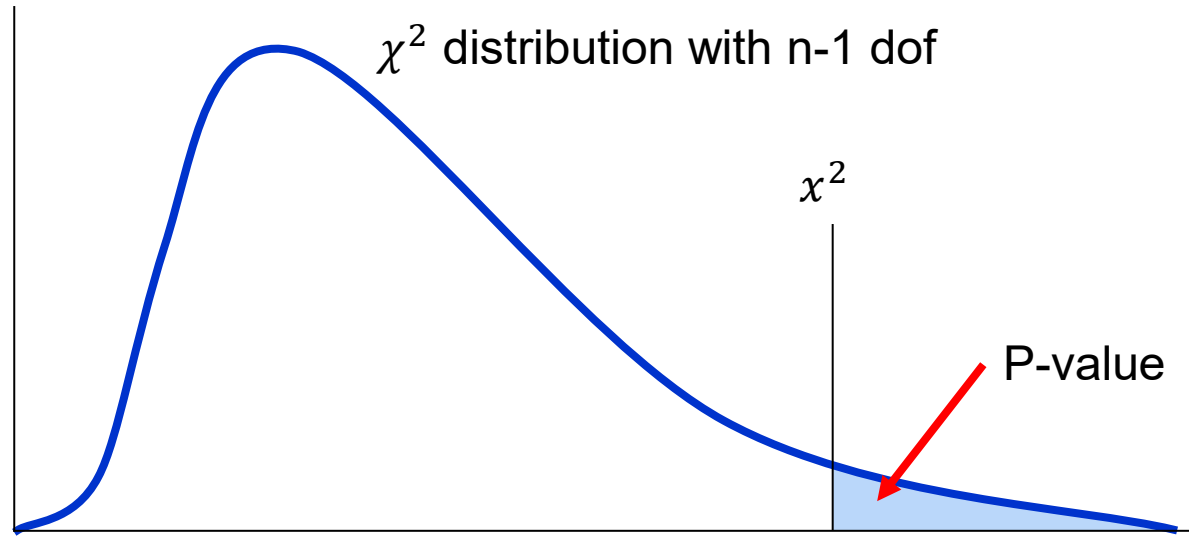
- χ^2 test for a variance compared to a target using a χ^2 test statistic ν
- $H_0: \sigma = 1.0 \text{ MPa}$
- $H_A: \sigma > 1.0 \text{ MPa}$

$$\nu = \frac{\sigma_{\text{target}}^2}{s^2}$$



χ^2 Test Statistic and p-value

$$\chi^2 = \frac{s^2}{\sigma^2}$$



Hypothesis Testing Example: Steps 3 – 5

Step 3: Risk Levels

- $\alpha = 0.05$
- $\beta = 0.20$

Step 4: Determine sample size

- We'll come back to this step later. For now use $n \geq 10$.

Step 5: Randomly sample and measure parts

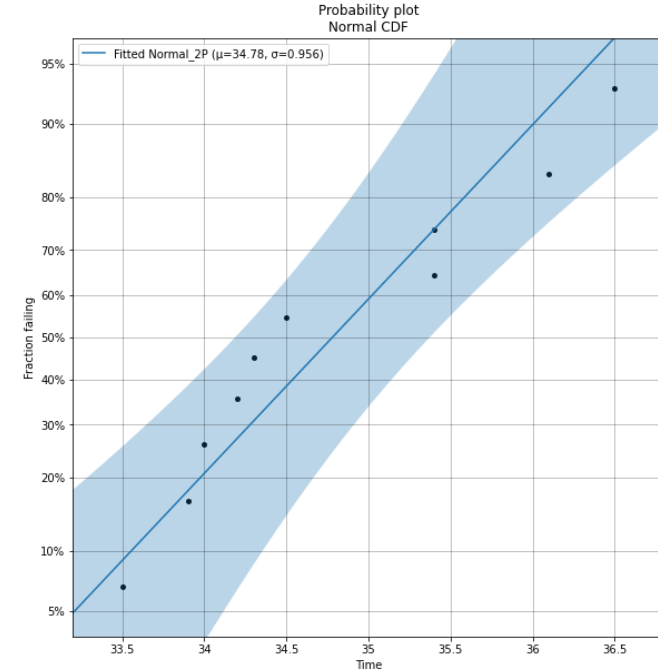
36.1
35.4
34.2
33.9
33.5
34.5
35.4
34.3
34.0
36.5



Test a Single Variance vs. a Target Value

Step 6: Check for the conditions of the test:

- Is the data normal?



Anderson Darling AD=0.396
p-value=0.300



Test a Single Variance vs. a Target Value

Steps 7-8: Do the test

Minitab: **Stat > Basic Stats > 1 Sample Variance**

Python: `mqr.inference.variance.test_1sample`

Test

Null hypothesis $H_0: \sigma = 1$

Alternative hypothesis $H_1: \sigma > 1$

Method	Test		
	Statistic	DF	P-Value
Bonett	—	—	0.482
Chi-Square	9.14	9	0.425

Our Conclusion Statement

Step 9: Statistical Decision: Do not reject H_0

- Since $p = 0.58$ is not low, we do not have sufficient statistical evidence to infer the sample comes from a population whose standard deviation is different than 1.0.

Step 10: Decision: Practically, we do not reject the supplier's claim that " $\sigma \leq 1.0$ ".

- Again, does this not prove that $\sigma \leq 1.0$.
- However, we operate under the assumption that "not rejecting" is "proving" and accept that H_0 is true.

We "accept" the manufacturer's claim.



Hypothesis Testing Example: Step 6 - Python

Step 6: Check to ensure the data is normal

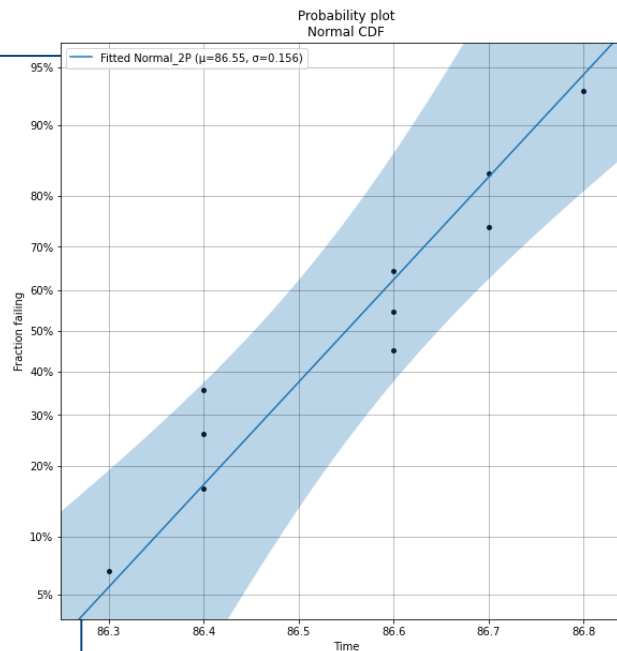
```
from reliability.Probability_plotting import
Normal_probability_plot
import matplotlib.pyplot as plt
import numpy as np
import mqr
import mqr.nbtools as nb

data = np.array([86.6, 86.7, 86.2, 86.6, 86.4,
                 86.3, 87.0, 86.4, 86.8, 86.7])

fig = Normal_probability_plot(failures=failures)
fig.set_size_inches(5, 4)
plot = nb.grab_figure(fig)

test = mqr.inference.dist.test_1sample(failures)

nb.hstack(plot, test)
```



Hypothesis Test non-normality

method Anderson-Darling

H₀ distribution == normal

H₁ distribution != normal

statistic 0.631534

p-value 0.0703465

H_0 : The data is normally distributed.
 $p=0.810$ is not low. Do not reject the null.

Hypothesis Testing Example: Step 6 - Minitab

Step 6: Check to ensure the data is normal

Stat > Basic Statistics > Normality Test

Normality Test

Variable: Data

Percentile Lines

☒ None

☐ At y values:

☐ At data values:

Tests for Normality

☒ Anderson-Darling

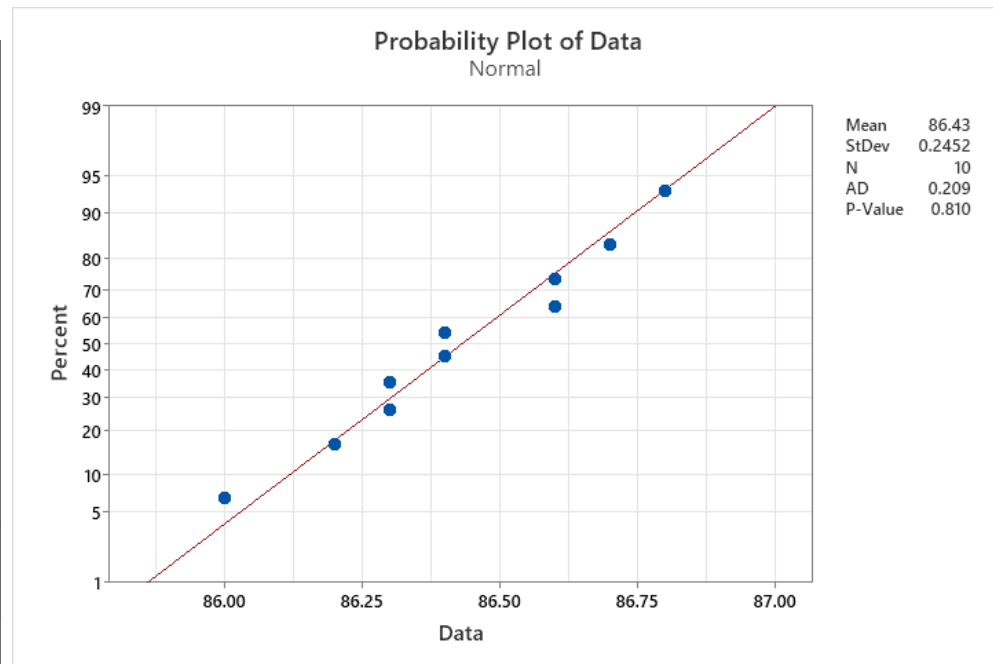
☐ Ryan-Joiner (Similar to Shapiro-Wilk)

☐ Kolmogorov-Smirnov

Select

Title:

Help OK Cancel



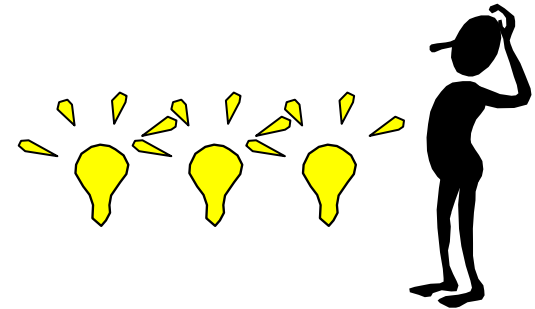
H_0 : The data is normally distributed.
 $p=0.810$ is not low. Do not reject the null.

T-test for a Mean vs Target Using Python

We do not have to do this manually for every set of data.

Statistical software has these calculations built in.

Lets re-do the lamp example using Minitab or Python statistical code libraries



Hypothesis Test: Steps 7-9 – Using Python

```
import scipy.stats as st

t, p = st.stats.ttest_1samp(data, 86.5, alternative='less')
print('t stat = %.3f' % (t))
print('p-value = %.3f' % (p))
```

```
t stat = -0.903
p-value = 0.195
```

Step 9: Make a decision. $p = 0.195$.

Since $p > 0.05$, do not reject H_0

Hypothesis Test: Steps 7-9 – Using Minitab

One-Sample t for the Mean

C1 Data

One or more samples, each in a column

Data

☒ Perform hypothesis test

Hypothesized mean: 86.5

Select

Options...

Graphs...

Help

OK

Cancel

One-Sample t: Options

Confidence level: 95.0

Alternative hypothesis: Mean < hypothesized mean

Help

OK

Cancel

Test

Null hypothesis	$H_0: \mu = 86.5$
Alternative hypothesis	$H_1: \mu < 86.5$
T-Value	P-Value
-0.90	0.195

Step 9: Make a decision.

$p = 0.195$.

Since $p > 0.05$, do not reject H_0

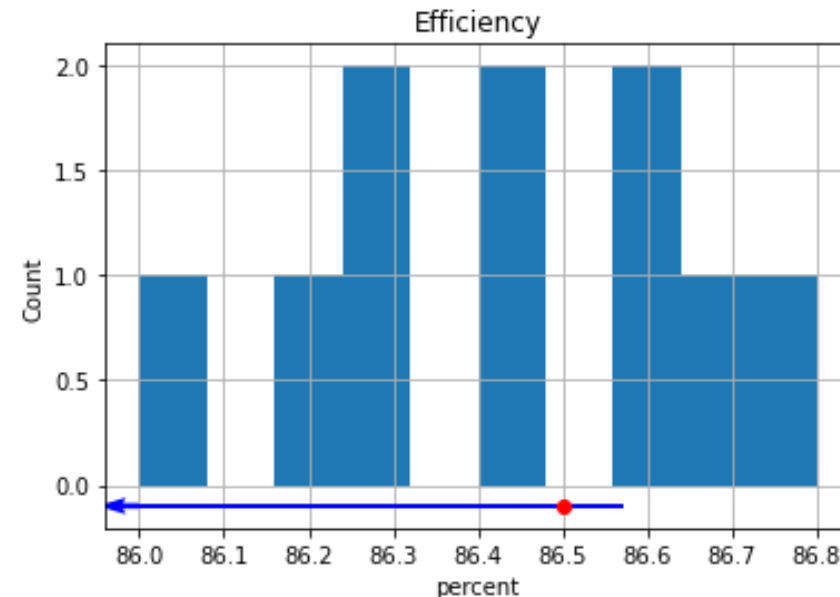
Graphical Interpretation: p-values and CIs

We can also graph the histogram of the data, show the null hypothesis target, and the one sided confidence interval. When the target is within the confidence interval, the p-value will be less than α and there is no difference.

```
import pandas as pd
import statsmodels.stats.api as sm

smData = sm.DescrStatsW(data)
xCI = smData.tconfint_mean(alpha=0.05,
                           alternative='smaller')

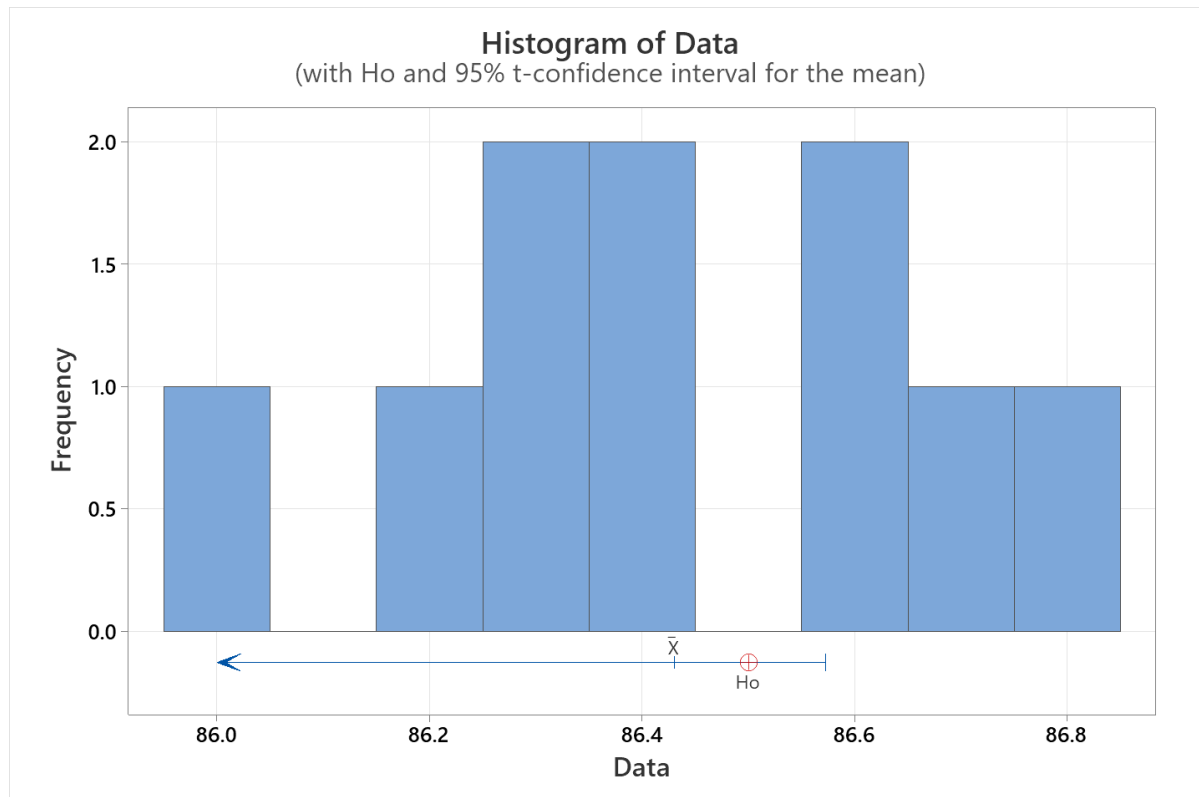
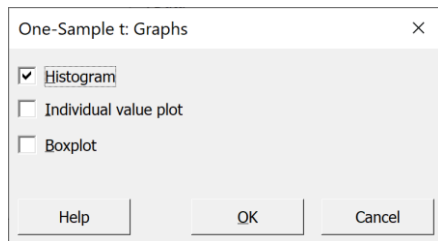
dfdata = pd.DataFrame(data)
dfdata.hist()
plt.title('Efficiency')
plt.xlabel('percent')
plt.ylabel('Count')
plt.hlines(y = -0.1, xmin=data.min(),
          xmax = xCI[1], linewidth=2, color='b',
          linestyle = '-')
plt.quiver(data.min(), -0.1, -1, 0, color='b')
plt.plot(86.5,-0.1,'ro')
```



xCI = (-inf, 86.572123730754)

Graphical Interpretation: p-values and CIs

We can also graph the histogram of the data, show the null hypothesis target, and the one sided confidence interval. When the target is within the confidence interval, the p-value will be less than α and there is no difference.



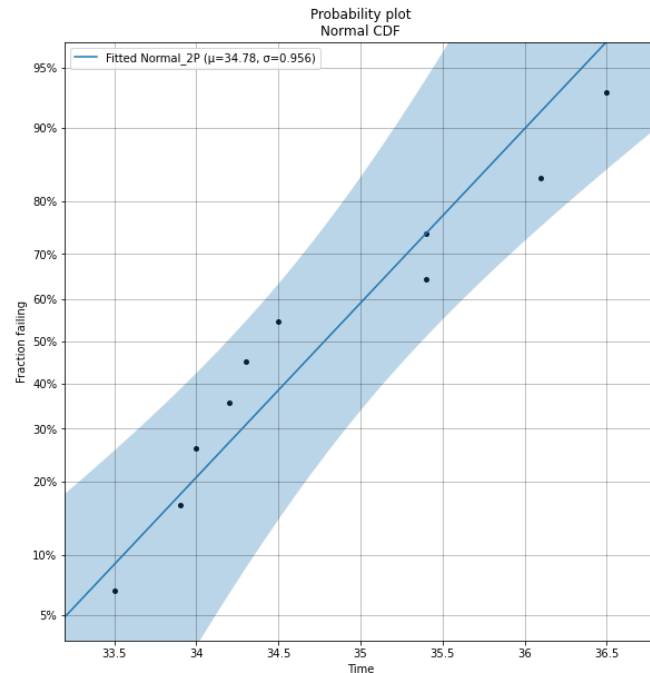
Test a Single Variance vs. a Target Value

Step 6: Check for the conditions of the test:

- Is the data normal?

```
data = np.array([36.1, 35.4, 34.2, 33.9,  
                 33.5, 34.5, 35.4, 34.3, 34.0, 36.5])  
ad2, pval = normal_ad(data)  
print('Anderson Darling AD=%.3f' % (ad2))  
print('p-value=%.3f' % (pval))
```

```
Normal_probability_plot(failures=data)  
plt.legend()  
plt.show()
```

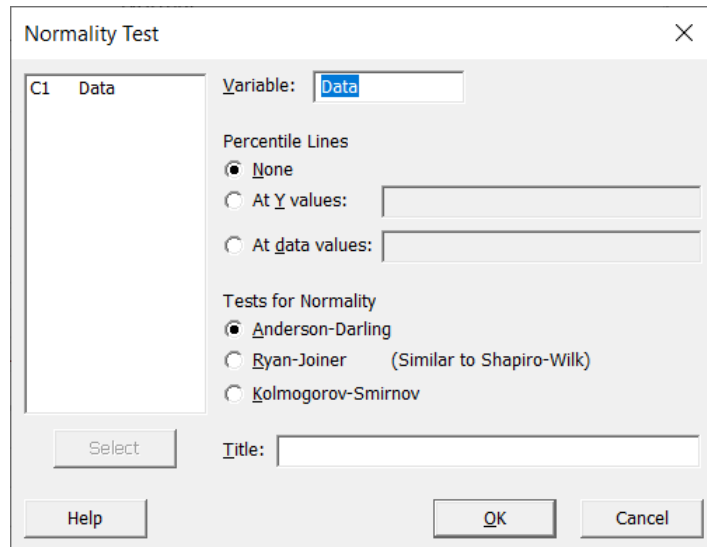


Anderson Darling AD=0.396
p-value=0.300

Test a Single Variance vs. a Target Value

Step 6: Check for the conditions of the test:

- Is the data normal?



Normality Test

Variable: Data

Percentile Lines

- ☒ None
- ☐ At Y values:
- ☐ At data values:

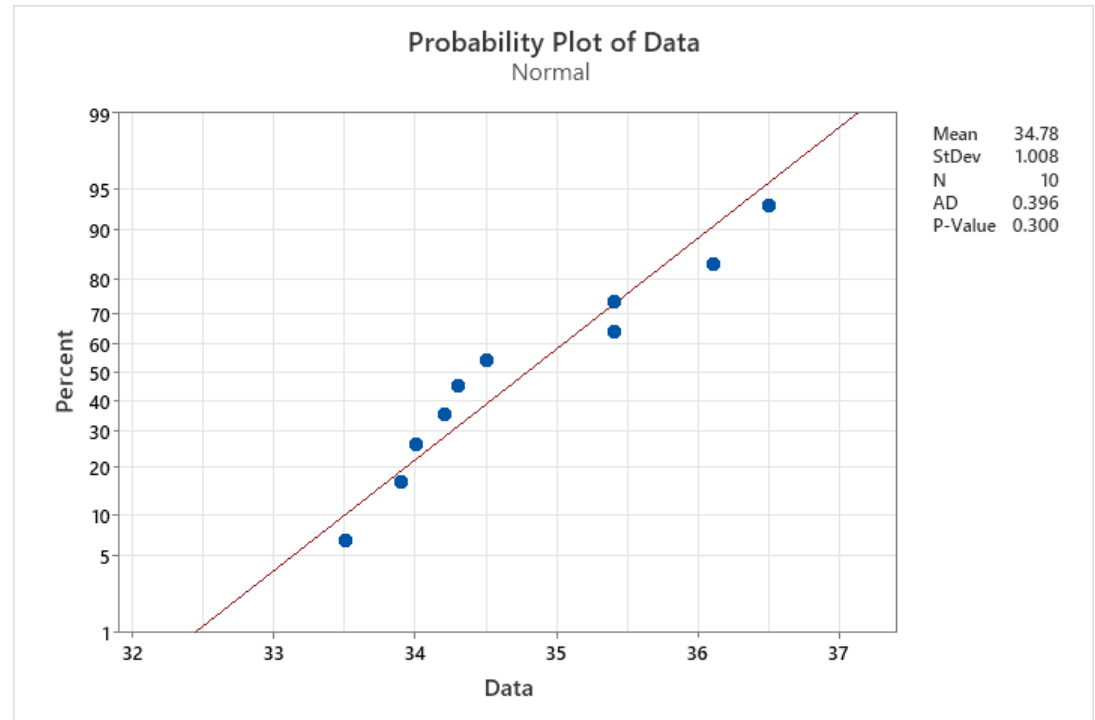
Tests for Normality

- ☒ Anderson-Darling
- ☐ Ryan-Joiner (Similar to Shapiro-Wilk)
- ☐ Kolmogorov-Smirnov

Select

Title:

Help OK Cancel



Test a Single Variance vs. a Target Value – Python

Steps 7-8: Do the test

The run it.

```

hyp = 1.0
alt = 'greater'

mqr.infer_variance(
    data, H0_var=hyp, alternative=alt)

```

Hypothesis Test
variance

method	chi2
H₀	var(x) == 1
H₁	var(x) > 1
statistic	1.01511
p-value	0.424816

Test a Single Variance vs. a Target Value – Minitab

Steps 7-8: Do the test

Stat > Basic Stats > 1 Sample Variance

Th

One-Sample Variance

One or more samples, each in a column

Data

☒ Perform hypothesis test

Hypothesized standard deviation Value: 1

Select Options... Help OK Cancel

One-Sample Variance: Options

Confidence level: 95.0

Alternative hypothesis: Standard deviation > hypothesized standard deviation

Help OK Cancel

Test

Null hypothesis $H_0: \sigma = 1$

Alternative hypothesis $H_1: \sigma > 1$

Method	Test		
	Statistic	DF	P-Value
Bonett	—	—	0.482
Chi-Square	9.14	9	0.425

Parametric Hypothesis Tests

Test	Purpose	Null
Anderson-Darling test Kolmogorov-Smirnov test	Data follows a distribution	follows
One sample t-test	Mean vs a target	$\mu = \mu_0$
One sample χ^2 test	Variance vs a target	$\sigma^2 = \sigma_0^2$
Two sample F test Barlett's test Levene's test	Two variances	$\sigma_1^2 = \sigma_2^2$
Two sample t-test	Two means	$\mu_1 = \mu_2$
Paired t-test	Two means of paired data	$\mu_1 = \mu_2$
One sample proportion test	Proportion vs a target	$\pi = \pi_0$
Two sample proportion test	Two Proportions	$\pi_1 = \pi_2$
One sample rate test	Rate vs a target	$\rho = \rho_0$
Two sample rate test	Two rates	$\rho_1 = \rho_2$

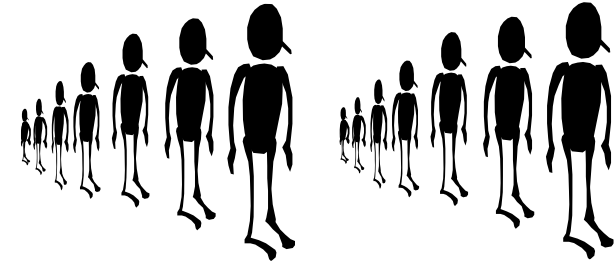


Test Two or More Independent Variables

Consider the question are two populations the same, or different?

At least two statistical questions can be posed

- Are the means different?
- Are the variances different?



Test Two or More Independent Variables

A motor has exhibited excess axial play. It was proposed to stake the plates in assembly.
10 randomly selected motors were staked and measured.
Another 10 randomly selected motors were not staked and measured.

Question:

Is axial play variance affected by staking or not?

Should we do the staking?

Step 1: State the problem.

Does staking affect the axial play variance?



Test for Two Independent Sample Variances

Step 2: Hypothesis

- 2 sample F-test: 2 independent random samples with and without staking
- Makes use of the F-statistic of two random samples $F = \frac{s_1^2}{s_2^2}$
- $H_0: \sigma_{\text{with}} = \sigma_{\text{without}}$ (actually $\frac{\sigma_1^2}{\sigma_2^2} = 1$)
- $H_A: \sigma_{\text{with}} \neq \sigma_{\text{without}}$ (actually $\frac{\sigma_1^2}{\sigma_2^2} \neq 1$)

Step 3: State risk levels

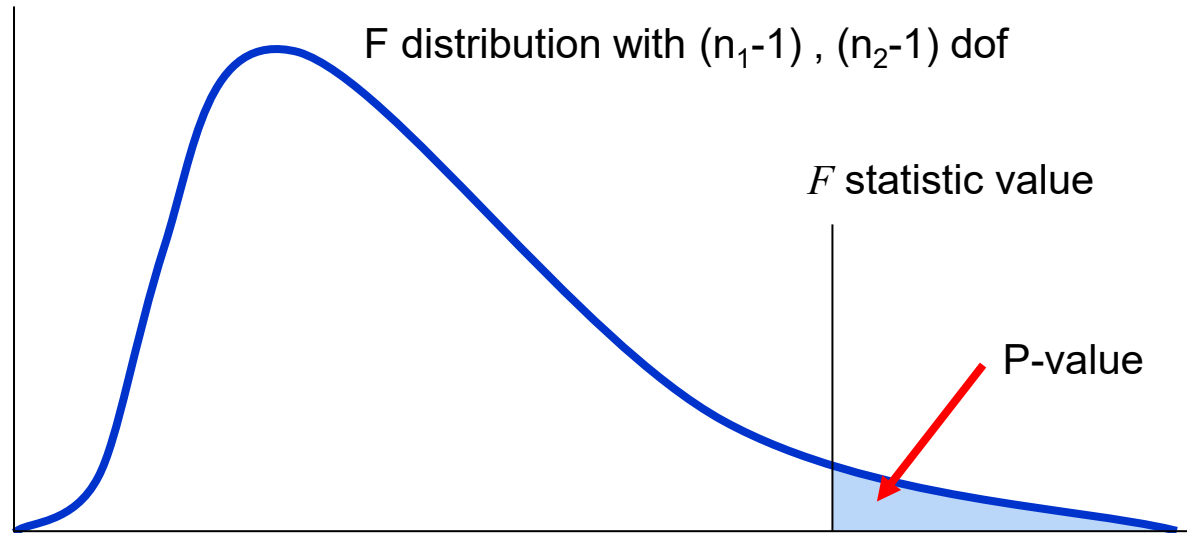
- $\alpha = 0.05$
- $\beta = 0.05$



Test for Two Independent Sample Variances

2 sample F-test

$$F = \frac{s_1^2}{s_2^2}$$



Hypothesis Testing Example: Steps 5 – 7

Step 4: Determine sample size

- We'll come back to this step later. For now use $n \geq 10$.

Step 5: Randomly sample parts and measure the data

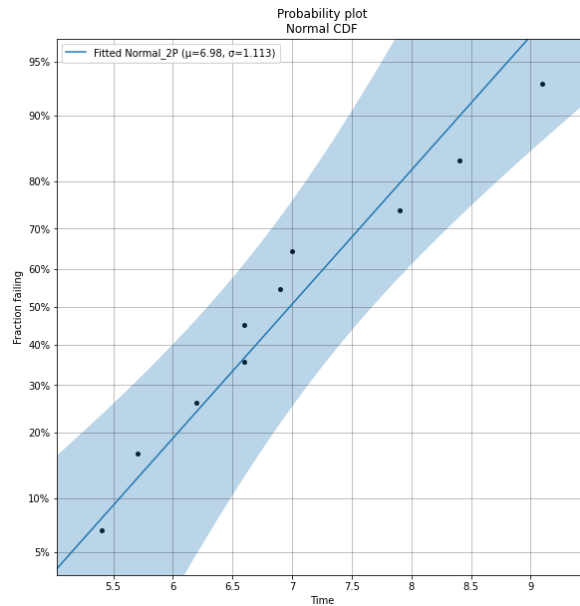
part	staked (microns)	unstaked (microns)
1	5.4	6.3
2	5.7	5.6
3	7.0	6.5
4	9.1	6.4
5	7.9	6.3
6	6.9	4.3
7	6.6	5.9
8	6.2	7.2
9	6.6	5.9
10	8.4	5.2



Test for Two Independent Sample Variances

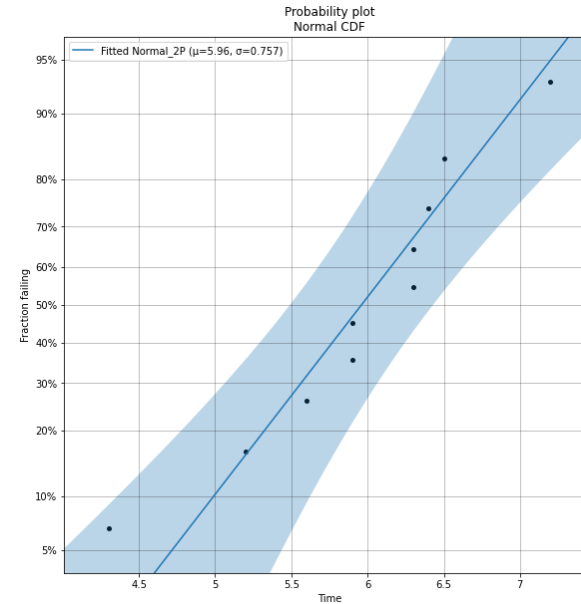
Step 6: Check for the conditions of the test: Is the data normal?

Staked



Anderson Darling AD2=0.259
p-value=0.632

Unstaked



Anderson Darling AD2=0.331
p-value=0.441



Test for Two Independent Sample Variances – Minitab

Step 7-8: Run the statistical test:

Minitab: **Statistics > Basic Statistics > Two Sample Variance**, options “Assume Normality”

Python: `f_test(x, y, alt)` :

Test

Null hypothesis $H_0: \sigma_1 / \sigma_2 = 1$

Alternative hypothesis $H_1: \sigma_1 / \sigma_2 < 1$

Significance level $\alpha = 0.05$

Method	Test			
	Statistic	DF1	DF2	P-Value
F	2.17	9	9	0.867

$p = 0.87 > 0.05$: p is not low.





Test for Two Independent Sample Variances

The F-test for equal variances, while easy to code and grasp, is known to be strongly dependent on the data being normally distributed.

Two other tests are more commonly used, the Bartlett and Levene test.

Bartlett's test for equal variances works well.

Minitab: **Statistics > Basic Statistics > Two Sample Variance**

Python: `=bartlett(data1, data2)`

Levene's test for equal variances is distribution free, though less statistically powerful.

Minitab: **Statistics > Basic Statistics > Two Sample Variance**

Python: `=levne(data1, data2)`

Test for Two Independent Sample Variances

Step 9: Statistical Decision: Do not reject H_0

- Since $p = 0.87 > 0.05$, we do not have sufficient statistical evidence to infer the sample comes from populations whose standard deviations are different.

Step 10: Decision: Practically, we do not reject the claim that $\sigma_1 = \sigma_2$.

We accept that staking had no impact on the variation.



Parametric Hypothesis Tests

Test	Purpose	Null
Anderson-Darling test Kolmogorov-Smirnov test	Data follows a distribution	follows
One sample t-test	Mean vs a target	$\mu = \mu_0$
One sample χ^2 test	Variance vs a target	$\sigma^2 = \sigma_0^2$
Two sample F test Barlett's test Levene's test	Two variances	$\sigma_1^2 = \sigma_2^2$
Two sample t-test	Two means	$\mu_1 = \mu_2$
Paired t-test	Two means of paired data	$\mu_1 = \mu_2$
One sample proportion test	Proportion vs a target	$\pi = \pi_0$
Two sample proportion test	Two Proportions	$\pi_1 = \pi_2$
One sample rate test	Rate vs a target	$\rho = \rho_0$
Two sample rate test	Two rates	$\rho_1 = \rho_2$



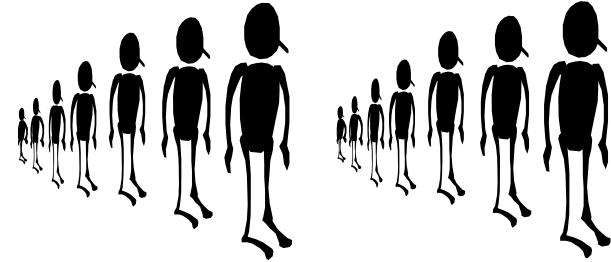
Test Two or More Independent Variables

Consider again the question are two populations the same, or different?

At least two statistical questions can be posed

- Are the means different?
- Are the variances different?

We will now address the former.



Test Two or More Independent Means

A motor has exhibited excess axial play. It was proposed to stake the plates in assembly.
10 randomly selected motors were staked and measured.
Another 10 randomly selected motors were not staked and measured.

Question:

Is axial play mean affected by staking or not?

Should we do the staking?

Step 1: State the problem.

Does staking affect the axial play mean?



Test for Two Independent Sample Means

Step 2: Hypothesis

- 2 sample t-test, 2-sided, using a t-statistic
- $H_0: \mu_{\text{with}} = \mu_{\text{without}}$
- $H_A: \mu_{\text{with}} \neq \mu_{\text{without}}$

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

Step 3: State risk levels

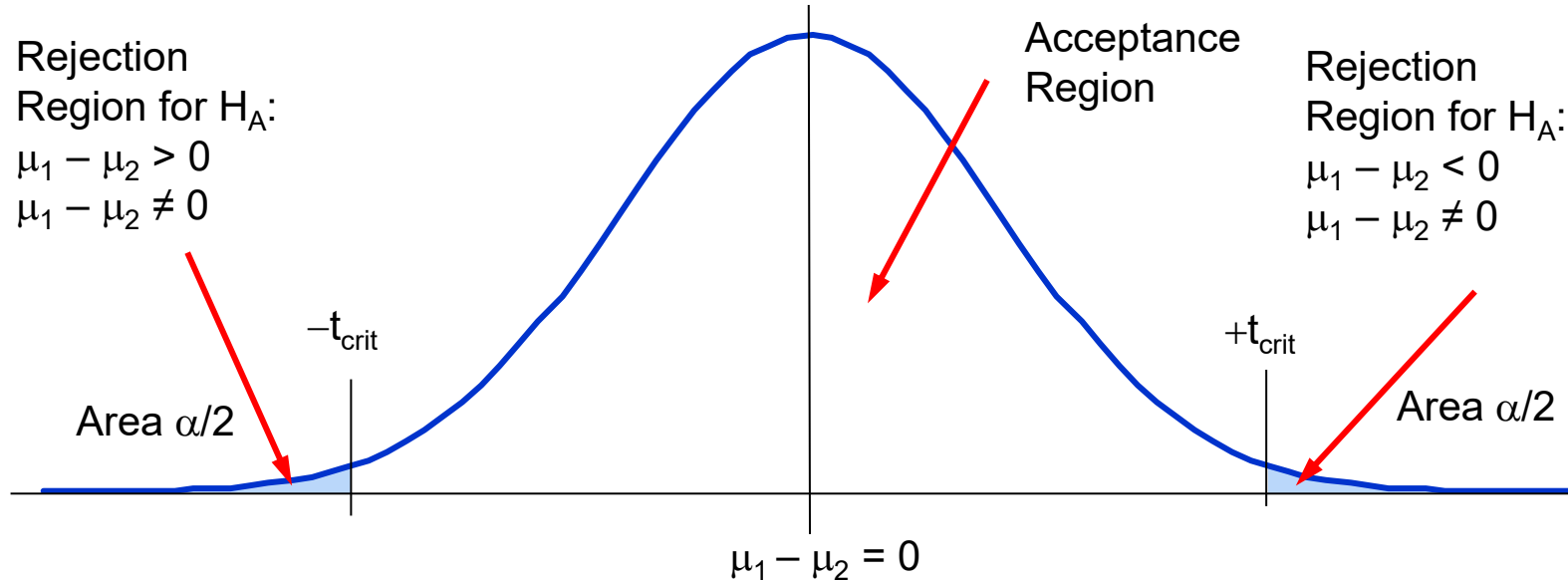
- $\alpha = 0.05$
- $\beta = 0.05$



Test for Two Independent Sample Means

2 sample t-test, 2-sided

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$



Hypothesis Testing Example: Steps 5 – 7

Step 4: Determine sample size

- We'll come back to this step later. For now use $n \geq 10$.

Step 5: Randomly sample parts and measure the data

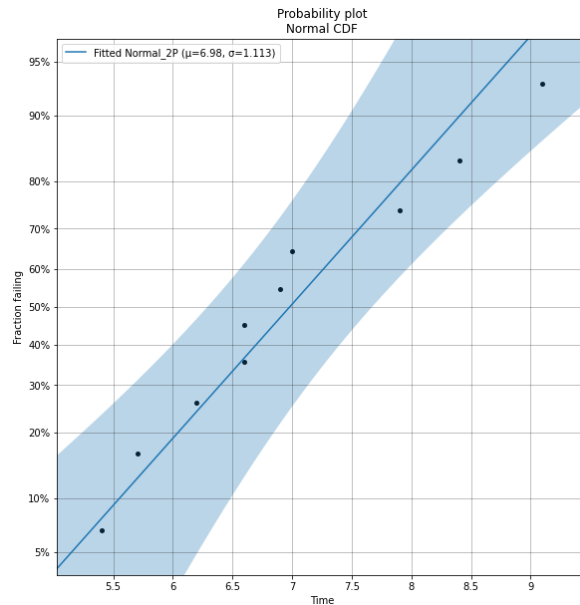
part	staked (microns)	unstaked (microns)
1	5.4	4.3
2	5.7	4.8
3	7.0	5.9
4	9.1	7.9
5	7.9	6.8
6	6.9	6.1
7	6.6	5.5
8	6.2	5.0
9	6.6	5.7
10	8.4	7.2



Test for Two Independent Sample Means

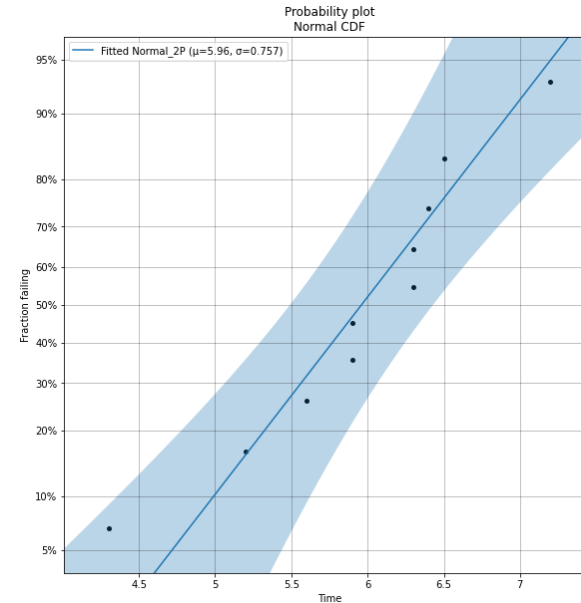
Step 6: Check for the conditions of the test: Is the data normal?

Staked



Anderson Darling AD2=0.259
p-value=0.632

Unstaked



Anderson Darling AD2=0.331
p-value=0.441



Test for Two Independent Sample Means

Step 6: Check for the conditions of the test:

Are the variances equal?

- There are two forms of this t-test, one assumes equal variances, the other does not. You can use either. Assuming equal variances provides more “power” that we will discuss later.

```
f ratio = 1.0954400848356305  
P-value = 0.44710715953396263
```

From the previous example, we accepted the variances were equal.



Test for Two Independent Sample Means - Minitab

Steps 7-8: Run the test

Minitab: **Statistics > Basic Statistics > Two-Sample t-test for the Mean**

Python: `mqr.inference.mean.test_2sample`

Test

Null hypothesis $H_0: \mu_1 - \mu_2 = 0$

Alternative hypothesis $H_1: \mu_1 - \mu_2 \neq 0$

T-Value	DF	P-Value
----------------	-----------	----------------

2.07	17	0.055
------	----	-------

$P = 0.055 > 0.05$: p is not low.

Test for Two Independent Sample Means

Step 9: Statistical Decision: Do not reject H_0

- Since $p > 0.05$, we do not have sufficient statistical evidence to infer the samples have different means.

Step 10: Decision: Practically, we do not reject the claim that $\mu_1 = \mu_2$.

- Does this prove that $\mu_1 = \mu_2$? No!
- However, we choose to accept the assumption that “not rejecting” is “proving” and accept that H_0 is true.

We “accept” that staking had no impact on the mean.



Hypothesis Testing Method Objectives

Test	Purpose	Null
Anderson-Darling test Kolmogorov-Smirnov test	Data follows a distribution	follows
One sample t-test	Mean vs a target	$\mu = \mu_0$
One sample χ^2 test	Variance vs a target	$\sigma^2 = \sigma_0^2$
Two sample F test Barlett's test Levene's test	Two variances	$\sigma_1^2 = \sigma_2^2$
Two sample t-test	Two means	$\mu_1 = \mu_2$
Paired t-test	Two means of paired data	$\mu_1 = \mu_2$
One sample proportion test	Proportion vs a target	$\pi = \pi_0$
Two sample proportion test	Two Proportions	$\pi_1 = \pi_2$
One sample rate test	Rate vs a target	$\rho = \rho_0$
Two sample rate test	Two rates	$\rho_1 = \rho_2$



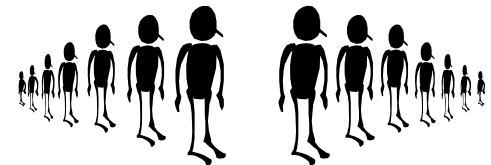
Paired Comparisons (The paired t-test)

Used when a one-to-one (dependent) correspondence exists between elements in the population

- Measurements before and after an operation
- Measurements with and without treatments on the same units
- Measurements by operator one or operator two

In paired comparisons we work with the differences of paired observations.

One can perform the one-sample t-test on these differences to determine if the mean is significantly different than a specified resolution delta (δ).



Test for Two Dependent Sample Means

A motor has exhibited excess axial play. It was proposed to stake the plates in assembly.
10 randomly selected motors were measured on axial play.
Then they were disassembled, staked, reassembled and again measured.

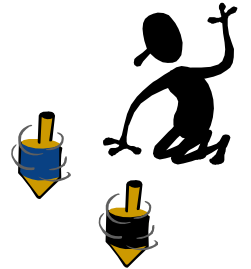
Question:

Is axial play mean affected by staking or not?

Should we do the staking?

Step 1: State the problem.

Does staking affect the axial play mean?



Paired t-test: The Concept

Convert paired data into a single data of the difference. Then do a one sample t-test on the difference of the population mean from zero.

Motor	before (microns)	after (microns)	difference (microns)
1	5.4	4.3	1.1
2	5.7	4.8	0.9
3	7.0	5.9	1.1
4	9.1	7.9	1.2
5	7.9	6.8	1.1
6	6.9	6.1	0.8
7	6.6	5.5	1.1
8	6.2	5.0	1.2
9	6.6	5.7	0.9
10	8.4	7.2	1.2

xbar	6.98	5.92	1.06
s	1.174	1.121	0.143

Test for Two Dependent Sample Means

Step 2: Hypothesis

- Paired t-test, 2-sided using a t-statistic
- $H_0: \mu_{\text{with}} = \mu_{\text{without}}$
- $H_A: \mu_{\text{with}} \neq \mu_{\text{without}}$

$$t = \frac{\bar{d}}{\frac{s_d}{\sqrt{n}}}$$

Step 3: State risk levels

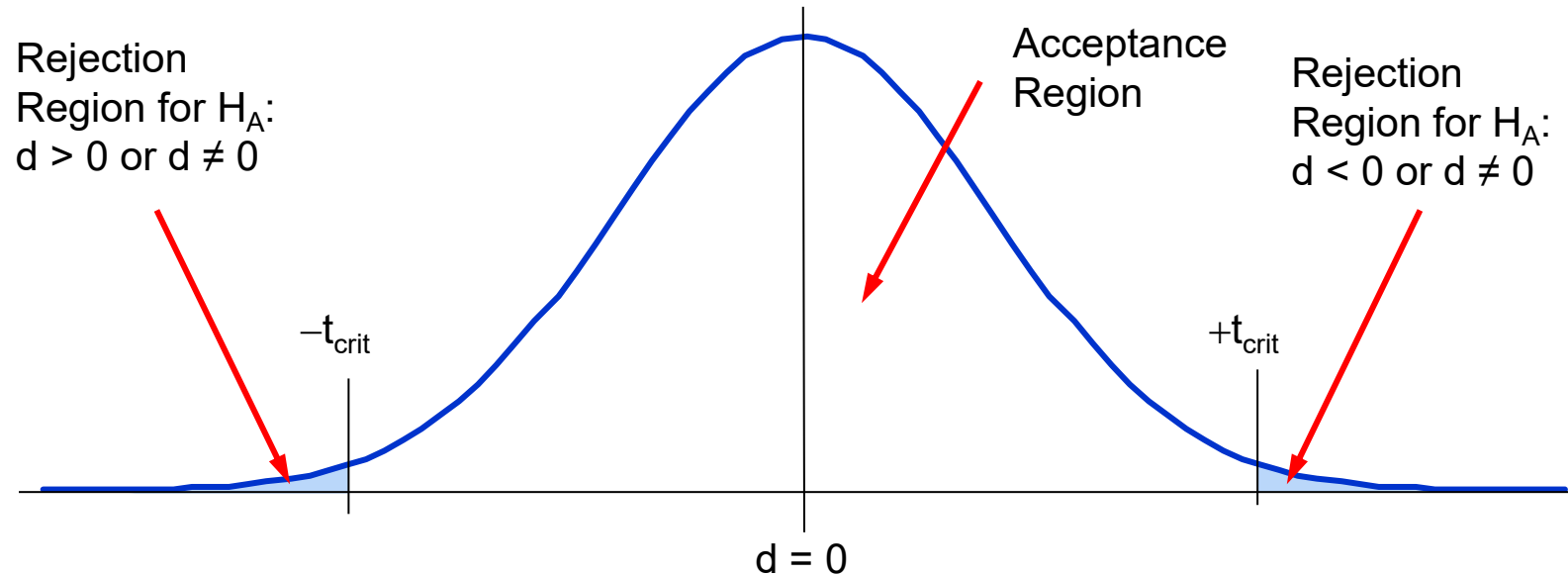
- $\alpha = 0.05$
- $\beta = 0.05$



Test for Two Dependent Sample Means

Paired t-test, 2-sided

$$t = \frac{\bar{d}}{\frac{s_d}{\sqrt{n}}}$$



Test for Two Dependent Sample Means

Step 4: Determine sample size

- We'll come back to this step later. For now use $n \geq 10$.

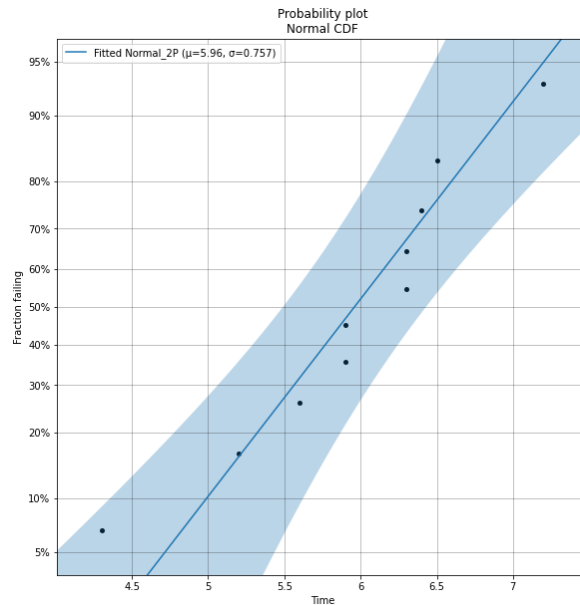
Step 5: Randomly sample parts and measure the data

part	before (microns)	after (microns)
1	5.4	4.3
2	5.7	4.8
3	7.0	5.9
4	9.1	7.9
5	7.9	6.8
6	6.9	6.1
7	6.6	5.5
8	6.2	5.0
9	6.6	5.7
10	8.4	7.2

Test for Two Independent Sample Means

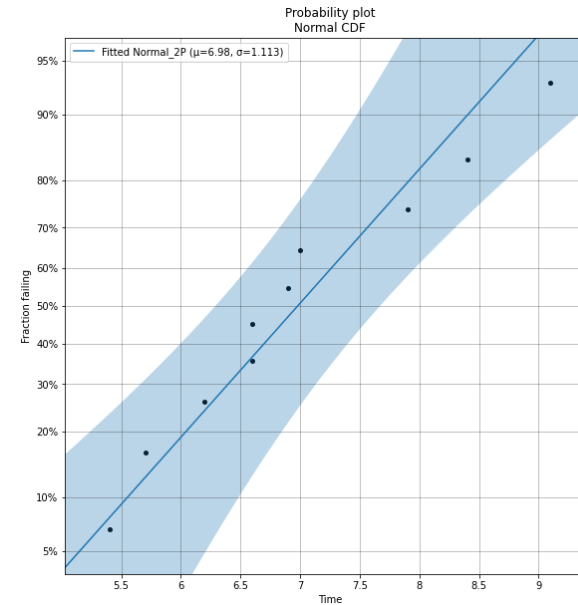
Step 6: Check for the conditions of the test: Is the data normal?

Before – Unstaked



Anderson Darling AD2=0.331
p-value=0.441

After – Staked



Anderson Darling AD2=0.259
p-value=0.632

Test for Two Dependent Sample Means

Steps 7-8: Run the test

Minitab: **Statistics > Basic Statistics > Paired t-test for the Mean**

Python: `ttest_rel(data1, data2, alternative)`

Test

Null hypothesis $H_0: \mu_{\text{difference}} = 0$

Alternative hypothesis $H_1: \mu_{\text{difference}} \neq 0$

T-Value	P-Value
----------------	----------------

23.44	0.000
-------	-------

$P = 0.000 < 0.05$: p is low.

Our Conclusion Statement

Step 9: Statistical Decision: Reject H_0

- Since $p = 0.000 < 0.05$, we do have sufficient statistical evidence to infer the samples have different means, to 95% confidence.

Step 10: Decision:

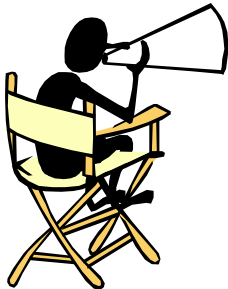
Practically, we reject the claim that $\mu_{\text{without}} = \mu_{\text{with}}$. We reject that staking had no impact. We accept the alternative.

We accept that staking did have an impact on the mean.



Hypothesis Testing Method Objectives

Test	Purpose	Null
Anderson-Darling test Kolmogorov-Smirnov test	Data follows a distribution	follows
One sample t-test	Mean vs a target	$\mu = \mu_0$
One sample χ^2 test	Variance vs a target	$\sigma^2 = \sigma_0^2$
Two sample F test Barlett's test Levene's test	Two variances	$\sigma_1^2 = \sigma_2^2$
Two sample t-test	Two means	$\mu_1 = \mu_2$
Paired t-test	Two means of paired data	$\mu_1 = \mu_2$
One sample proportion test	Proportion vs a target	$\pi = \pi_0$
Two sample proportion test	Two Proportions	$\pi_1 = \pi_2$
One sample rate test	Rate vs a target	$\rho = \rho_0$
Two sample rate test	Two rates	$\rho_1 = \rho_2$





Tests for One Proportion

One proportion tests: Is the population proportion equal to a “standard”

Suppose a sample of 100 parts were randomly sampled and inspected from today's production, and 3 defectives were observed. Is the yield less than 99%?

Hypothesis:

- Proportion test, 1-sided. The proportion 'standard' to test against is 1%, the maximum allowed scrap rate.
- $H_0: \pi = 0.01$
- $H_A: \pi > 0.01$



Tests for One Proportion

Run the test

Minitab: **Statistics > Basic Statistics > One Sample Proportion Test**

Python: `mqr.inference.proportion.test_1sample`

$p = 0.08 > 0.05$: p is not low.

Since $p = 0.08 > 0.05$, we do not have

sufficient statistical evidence to infer the population yield is not 99%, to 95% confidence.

Practically, we do not reject the claim that $\pi = 1\%$. Without more evidence, we accept that yield is 99%.

What happens if we had observed 30 in 1000?

Test

Null hypothesis $H_0: p = 0.01$

Alternative hypothesis $H_1: p > 0.01$

P-Value

0.079



Tests for Two Proportions

Two proportion tests: Are the population proportions from two samples equal

Suppose a sample of 100 parts were randomly sampled and inspected from today's production, and 3 defectives were observed. Suppose yesterday a sample of 100 parts were randomly sampled and inspected and 0 defectives were observed. Did the process change?

Step 2: Hypothesis

- Two Proportion test, 1-sided.
- $H_0: \pi_1 = \pi_2$
- $H_A: \pi_1 \neq \pi_2$



Tests for Two Proportions

Run the test

Minitab: **Statistics > Basic Statistics > Two Sample Proportion Test**

Python: `mqr.inference.proportion.test_2sample`

Test

Null hypothesis $H_0: p_1 - p_2 = 0$

Alternative hypothesis $H_1: p_1 - p_2 \neq 0$

Method	Z-Value	P-Value
--------	---------	---------

Normal approximation	-1.76	0.079
----------------------	-------	-------

Fisher's exact		0.246
----------------	--	-------

The normal approximation may be inaccurate for small samples.

$p = 0.08 > 0.05$: p is not low. We do not have sufficient statistical evidence to not infer the population yields are the same. Practically, we do not reject the claim that $\pi_1 = \pi_2$.

Without more evidence, we “accept” that yields are the same and has not changed.

What happens if we had observed 0 in 1000 and 30 in 1000?

Test for Two Independent Sample Variances – Python

Step 7-8: Run the statistical test:

```
from scipy.stats import f as fdist

def f_test(x, y, alt):
    x = np.array(x)
    y = np.array(y)
    f = np.var(x, ddof=1)/np.var(y, ddof=1)
    dfn = x.size-1                #define degrees of freedom numerator
    dfd = y.size-1                #define degrees of freedom denominator
    if alt == "<":
        p = fdist.cdf(f, dfn, dfd) #find p-value of F test statistic
    elif alt == ">":
        p = 1-fdist.cdf(f, dfn, dfd) #find p-value of F test statistic
    else:
        p = 2.0*np.minimum(1-fdist.cdf(f, dfn, dfd),fdist.cdf(f, dfn, dfd))
    return f, p

data1 = np.array([5.4, 5.7, 7.0, 9.1, 7.9, 6.9, 6.6, 6.2, 6.6, 8.4])
data2 = np.array([4.3, 4.8, 5.9, 7.9, 6.8, 6.1, 5.5, 5.0, 5.7, 7.2])
alt = "<"    # one of "<", ">", or "!="

f, p = f_test(data1, data2, alt)
print('F-test for a less-than alternative hypothesis')
print('Test statistic = %.3f' % (f))
print('Sigma ratio = %.3f' % (data1.std(ddof=1)/data2.std(ddof=1)))
print('1-tailed p-value = %.3f' % (p))
```

Test for Two Independent Sample Variances – Python

Step 7-8: Run the statistical test:

```
F-test for a  $s^2_1 < s^2_2$  alternative hypothesis  
Test statistic = 2.166  
Sigma ratio = 1.472  
p-value = 0.867
```

$p = 0.87 > 0.05$: p is not low.



Test for Two Independent Sample Variances – Minitab

Step 7-8: Run the statistical test:

Two-Sample Variance

Each sample is in its own column

Sample 1: s1

Sample 2: s2

Select Options... Graphs... Results...

Help OK Cancel

Test

Null hypothesis $H_0: \sigma_1 / \sigma_2 = 1$

Alternative hypothesis $H_1: \sigma_1 / \sigma_2 < 1$

Significance level $\alpha = 0.05$

Method	Test Statistic	DF1	DF2	P-Value
F	2.17	9	9	0.867

$p = 0.87 > 0.05$: p is not low.



Test for Two Independent Sample Variances

Step 7-8: Run the statistical test:

Test

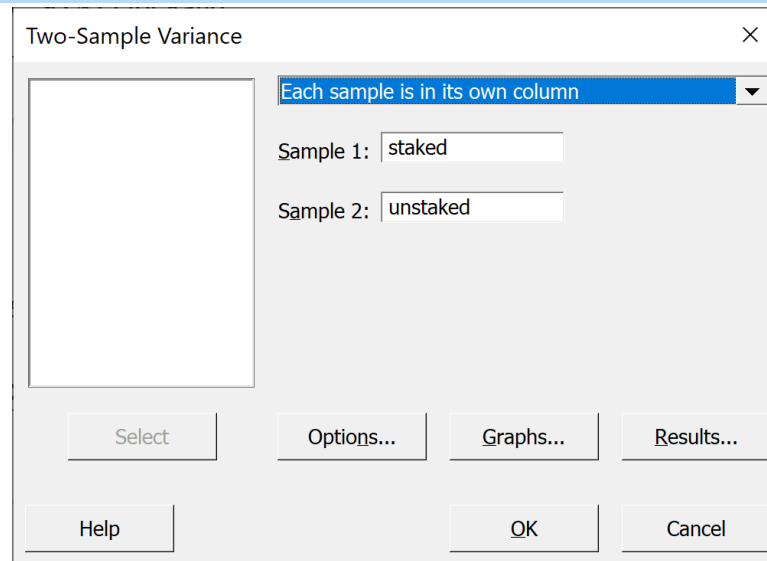
Null hypothesis $H_0: \sigma_1 / \sigma_2 = 1$

Alternative hypothesis $H_1: \sigma_1 / \sigma_2 \neq 1$

Significance level $\alpha = 0.05$

Method	Test Statistic	DF1	DF2	P-Value
Bonett	1.13	1		0.287
Levene	1.05	1	18	0.320

$p = 0.3 > 0.05$: p is not low.




Test for Two Independent Sample Means

Step 6: Check for the conditions of the test:

Are the variances equal?

- There are two forms of this t-test, one assumes equal variances, the other does not. You can use either. Assuming equal variances provides more “power” that we will discuss later.

```
f ratio = 1.0954400848356305  
P-value = 0.44710715953396263
```

From the previous example, we accepted the variances were equal.

Test for Two Independent Sample Means – Python

Steps 7-8: Run the test

```
import statsmodels.api as sm

data1 = np.array([5.4, 5.7, 7.0, 9.1, 7.9, 6.9, 6.6, 6.2, 6.6, 8.4])
data2 = np.array([4.3, 4.8, 5.9, 7.9, 6.8, 6.1, 5.5, 5.0, 5.7, 7.2])

print('Estimate of the difference:', np.average(data1) - np.average(data2))
tstat, p, df = sm.stats.ttest_ind(data1, data2, alternative='two-sided', usevar='pooled')
print('t stat = %.3f' % (tstat))
print('p-value = %.3f' % (p))
```

```
Estimate of the difference: 1.060
t ratio = 2.07
P-value = 0.055
```

$P = 0.055 > 0.05$: p is not low.

Test for Two Independent Sample Means - Minitab

Steps 7-8: Run the test

Two-Sample t for the Mean

Each sample is in its own column

Sample 1: S1

Sample 2: S2

Select Options... Graphs...

Help OK Cancel

Test

Null hypothesis $H_0: \mu_1 - \mu_2 = 0$

Alternative hypothesis $H_1: \mu_1 - \mu_2 \neq 0$

T-Value	DF	P-Value
2.07	17	0.055

$P = 0.055 > 0.05$: p is not low.

Test for Two Dependent Sample Means – Python

Steps 7-8: Run the test

```
from scipy.stats import ttest_rel

data1 = np.array([5.4, 5.7, 7.0, 9.1, 7.9, 6.9, 6.6, 6.2, 6.6, 8.4])
data2 = np.array([4.3, 4.8, 5.9, 7.9, 6.8, 6.1, 5.5, 5.0, 5.7, 7.2])

print('Estimate of the difference:', np.average(data1) - np.average(data2))
tstat, p = ttest_rel(data1, data2, alternative='two-sided')
print('t ratio =', tstat)
print('p-value = ', p)
```

```
Estimate of the difference: 1.06000000000000014
t ratio = 23.443
p-value = 0.000
```

$P = 0.000 < 0.05$: p is low.

Test for Two Dependent Sample Means - Minitab

Steps 7-8: Run the test

Paired t for the Mean

Each sample is in a column

Sample 1: staked

Sample 2: unstaked

Select Options... Graphs...

Help OK Cancel

Test

Null hypothesis $H_0: \mu_{\text{difference}} = 0$

Alternative hypothesis $H_1: \mu_{\text{difference}} \neq 0$

T-Value **P-Value**

23.44

0.000

$P = 0.000 < 0.05$: p is low.

Tests for One Proportion – Python

Run the test

```
import numpy as np, statsmodels.stats.api as sm

count = 3
nobs = 100

p = sm.binom_test(count, nobs, prop=0.01, alternative='larger')
print('p-value = %.3f' % (p))
```

```
0.07937320225218035
```

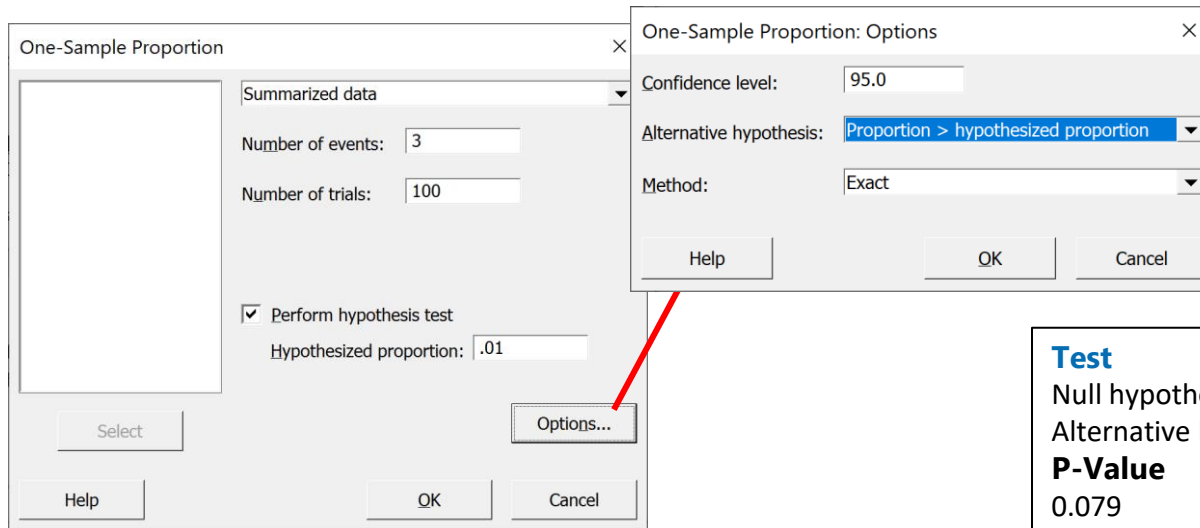
$p = 0.08 > 0.05$: p is not low. Since $p = 0.08 > 0.05$, we do not have sufficient statistical evidence to infer the population yield is not 99%, to 95% confidence.

Practically, we do not reject the claim that $\pi = 1\%$. Without more evidence, we accept that yield is 99%.

What happens if we had observed 30 in 1000?

Tests for One Proportion – Minitab

Run the test



The image shows two Minitab dialog boxes. The main 'One-Sample Proportion' dialog has 'Summarized data' selected, 'Number of events' set to 3, 'Number of trials' set to 100, and 'Perform hypothesis test' checked with a 'Hypothesized proportion' of .01. The 'Options...' button is highlighted with a red arrow. The 'One-Sample Proportion: Options' sub-dialog shows a 'Confidence level' of 95.0, 'Alternative hypothesis' set to 'Proportion > hypothesized proportion', and 'Method' set to 'Exact'.

Test

Null hypothesis $H_0: p = 0.01$

Alternative hypothesis $H_1: p > 0.01$

P-Value

0.079

$p = 0.08 > 0.05$: p is not low. Since $p = 0.08 > 0.05$, we do not have sufficient statistical evidence to infer the population yield is not 99%, to 95% confidence.

Practically, we do not reject the claim that $\pi = 1\%$. Without more evidence, we accept that yield is 99%.

What happens if we had observed 30 in 1000?

Tests for Two Proportions – Python

Run the test

```
import statsmodels.stats.api as sm
from statsmodels.stats.proportion import test_proportions_2indep

count1 = 0
nobs1 = 100
count2 = 3
nobs2 = 100

test_proportions_2indep(count1, nobs1, count2, nobs2,
                        alternative='two-sided')
```

```
statistic = -1.364576478442026
pvalue = 0.17238620178886022
```

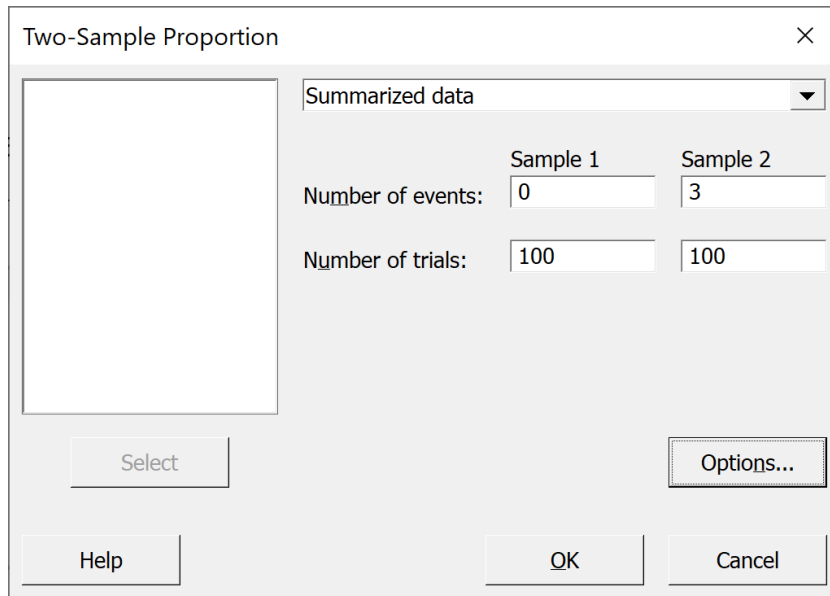
$p = 0.17 > 0.05$: p is not low. We do not have sufficient statistical evidence to not infer the population yields are the same. Practically, we do not reject the claim that $\pi_1 = \pi_2$.

Without more evidence, we “accept” that yields are the same and has not changed.

What happens if we had observed 0 in 1000 and 30 in 1000?

Tests for Two Proportions – Minitab

Run the test



Two-Sample Proportion

Summarized data

Number of events: Sample 1: 0, Sample 2: 3

Number of trials: 100, 100

Select Options... Help OK Cancel

Test

Null hypothesis $H_0: p_1 - p_2 = 0$

Alternative hypothesis $H_1: p_1 - p_2 \neq 0$

Method **Z-Value** **P-Value**

Normal approximation -1.76 0.079

Fisher's exact 0.246

The normal approximation may be inaccurate for small samples.

$p = 0.08 > 0.05$: p is not low. We do not have sufficient statistical evidence to not infer the population yields are the same. Practically, we do not reject the claim that $\pi_1 = \pi_2$.

Without more evidence, we “accept” that yields are the same and has not changed.

What happens if we had observed 0 in 1000 and 30 in 1000?



Overview

Test Errors

Statistical Inferencing

T-test

Parametric tests

Sample Sizes

Non-parametric tests



Sample Size Selection

Sensitivity: the size of the sample dictates the resolution or sensitivity of any comparisons that are made.

Risk Control: we can control the risks of making an incorrect conclusion from the data.

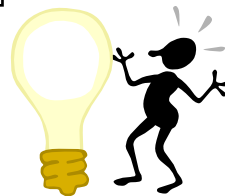
Systematic Approach: we can systematically and accurately determine an appropriate sample size, which is also useful in determining the number of replicates and repetitions needed in the planning and execution of designed experiments.

Some questions we will be able to answer:

- How many samples should be taken?
- Was the sample pulled large enough?
- Should more samples be taken just to be sure?

Understanding Statistical Decision Errors

		Reality	
		H_0 TRUE	H_0 FALSE
Decision	ACCEPT H_0	Correct Acceptance CONFIDENCE $1 - \alpha$	Type II Error Beta Risk β
	REJECT H_0	Type I Error Alpha Risk α	Correct Rejection POWER $1 - \beta$





Focus on the Critical Difference

Besides quantifying the α and β risks, next define the ***critical difference***, δ .

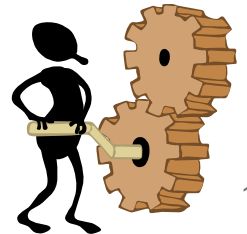
This difference is defined as the practical difference that the experimenter wants to be able to distinguish to say “it is different”. How small does this have to be able to be?

Smaller δ requires a larger sample size.

We assign the Type II error to the possibility of missing this critical difference. In this way, the alternative hypothesis can take on a specific meaning.

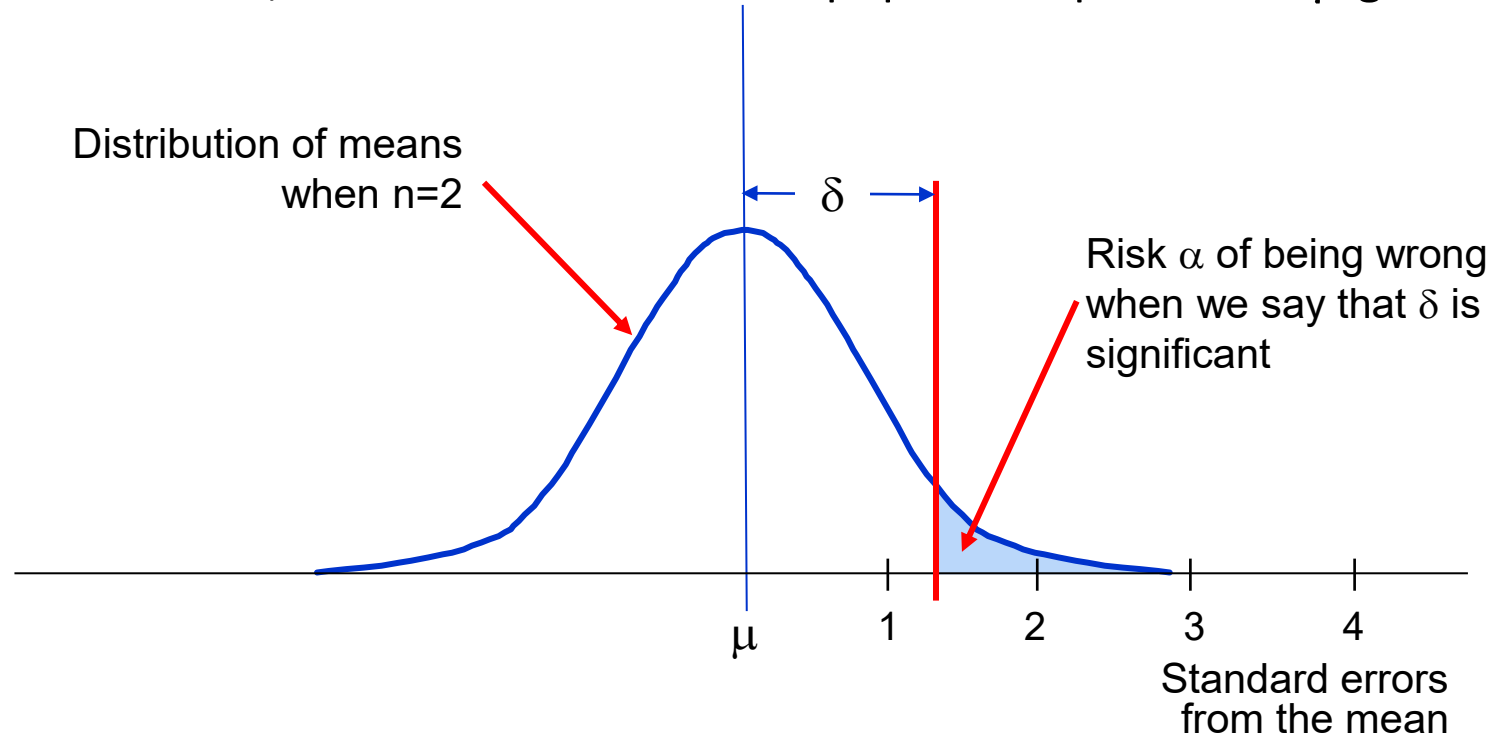
Sample Size Selection Method: Step by Step

1. Select alpha (α) level – typically 5%
Select beta (β) level – typically 5 – 20%
2. Establish the difference (δ) you would like to be able to detect in your experiment
3. Establish or estimate current process sigma (σ)
4. Use a sample size calculator to determine n
5. Develop the sampling plan



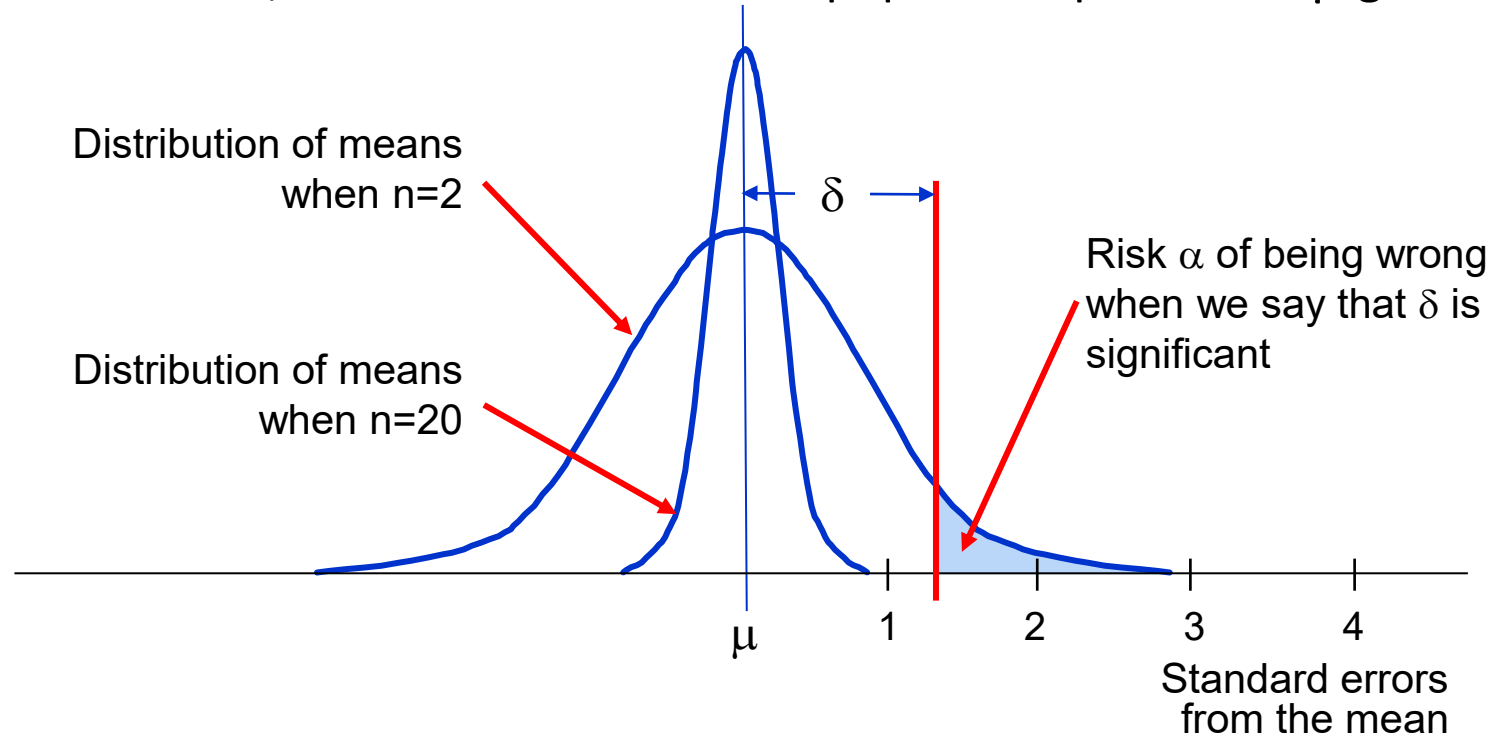
Relationship Between n , δ and Sigma

As sample size increases, the estimate of the true population parameter μ gets better



Relationship Between n , δ and Sigma

As sample size increases, the estimate of the true population parameter μ gets better





The δ/σ “Effect” Ratio

The ratio δ/σ represents the difference to detect measured in standard deviations.

- When the ratio is high, the difference can be easily detected.
- Sometimes called the *effect*

If variation in a process is high, it is more difficult to determine whether a difference is due to random chance, or whether there is truly a difference.

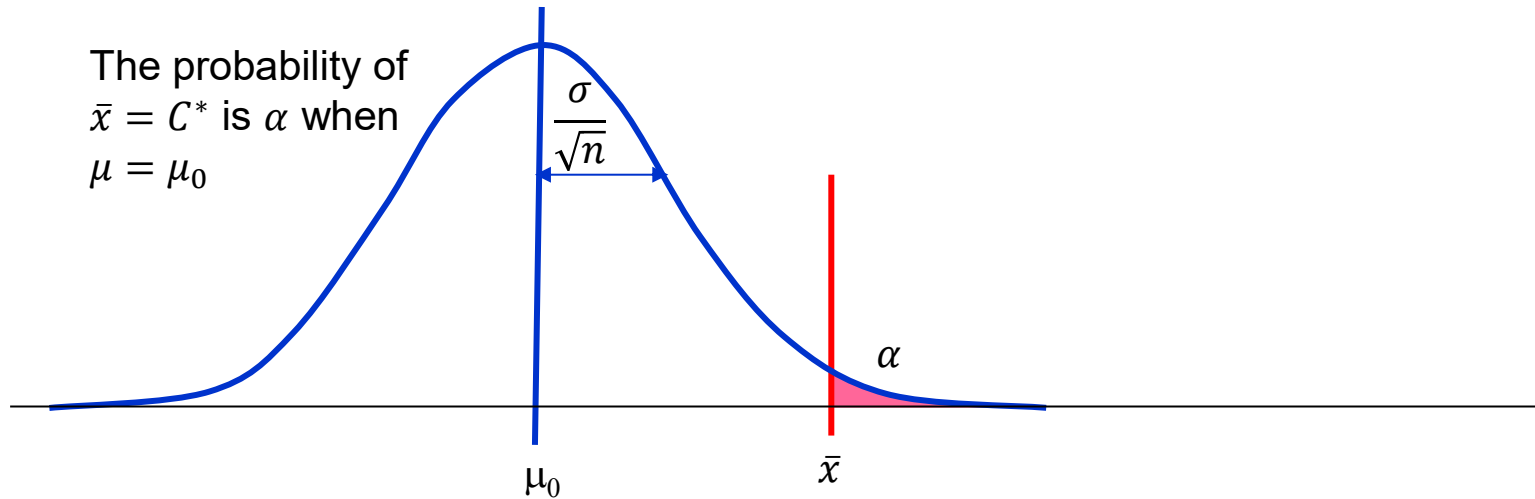
- More samples are needed to ferret out the truth.

As the number of sampled units we take increases, the theoretical standard deviation of our estimate of the mean decreases.

- We can reliably detect smaller and smaller differences.

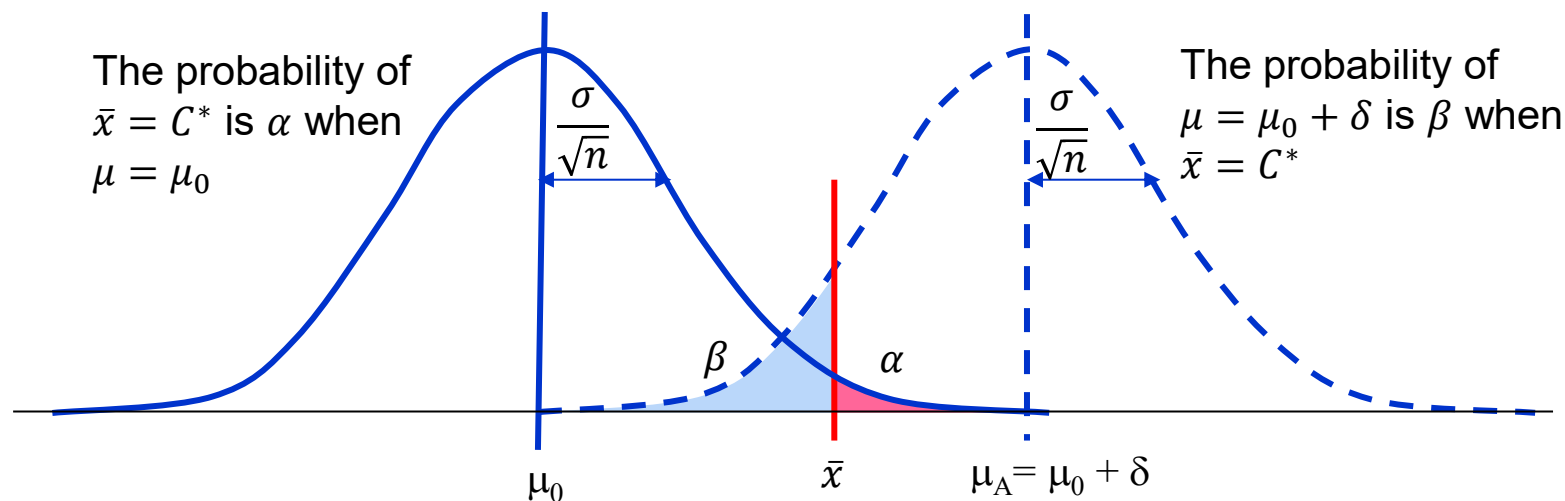
Sample size from α and β Risks

Consider a sample estimate $\bar{x}$, given a μ_0 target.
How much α and β risk is there?



Sample size from α and β Risks

Consider a sample estimate $\bar{x}$, given a μ_0 target.
How much α and β risk is there?



Sample size from α and β Risks

Given $\bar{x}$, the probability of $\mu = \mu_0$ is α , and

$$\bar{x} = \mu_0 + \frac{Z_{1-\alpha}\sigma}{\sqrt{n}}$$

Also, given $\bar{x}$ the probability of $\mu = \mu_0 + \delta$ is β , and

$$\bar{x} = \mu_0 + \delta - \frac{Z_{1-\beta}\sigma}{\sqrt{n}}$$

Setting these two equal to each other and solving for n :

$$n = \frac{(Z_{1-\alpha} + Z_{1-\beta})^2 \sigma^2}{\delta^2}$$

This is the **sample size equation**. Use it to compute the sample size for a t-test.



Choosing α , β and δ

Choosing values for α , β and δ is one of the most difficult tasks for an experimenter. Typically, one should assess the costs of making Type I and Type II errors:

Does the size of the difference (δ) justify the cost of the number of samples?

What will it cost to implement a change, when it may in fact be ineffective (α risk)?

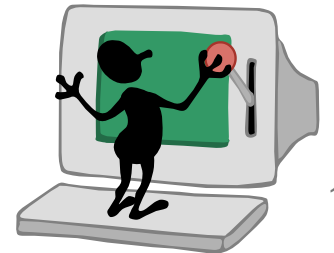
What will it cost to ignore a possible benefit or negative side effect (β risk)?

Calculating Sample Sizes

Sample sizes for a variety of circumstances can be calculated using the appropriate statistical distributions

- Given the values of α , β and δ .
- Given the historic σ^2 variance, or the historic scrap rate π_0 .

Software code makes these calculations easy.



Testing Mean Sample Size – Python

To determine the sample size for testing the mean:

Test	Purpose	Equation	Python StatsModels code
One sample Z-test	Comparison to a known mean, σ known	$n = \frac{(Z_{1-\alpha} + Z_{1-\beta})^2 \sigma^2}{\delta^2}$	<code>mqr.inference.mean.test_1sample</code>
One sample t-test	Comparison to a known mean, σ unknown	$n = \frac{(t_{1-\alpha} + t_{1-\beta})^2 s^2}{\delta^2}$	<code>mqr.inference.mean.test_1sample</code>
Two sample Z-test	Comparison of two unknown mean, σ known and equal	$n = \frac{2(Z_{1-\alpha} + Z_{1-\beta})^2 \sigma^2}{\delta^2}$	<code>mqr.inference.mean.test_2sample</code>
Two sample t-test	Comparison of two unknown mean, σ 's unknown	$n = \frac{2(t_{1-\alpha} + t_{1-\beta})^2 s^2}{\delta^2}$	<code>mqr.inference.mean.test_2sample</code>
Paired t-test	Two means of paired data with differences d_j	$n = \frac{(t_{1-\alpha} + t_{1-\beta})^2 s_d^2}{\delta^2}$	<code>mqr.inference.mean.test_paired</code>

These formulas are for one-sided tests. Replace α by $\alpha/2$ for a two sided test.

Testing Mean Sample Size – Minitab

To determine the sample size for testing the mean:

Test	Purpose	Equation	Minitab
One sample Z-test	Comparison to a known mean, σ known	$n = \frac{(Z_{1-\alpha} + Z_{1-\beta})^2 \sigma^2}{\delta^2}$	Stat > Power and Sample Size > 1-sample Z...
One sample t-test	Comparison to a known mean, σ unknown	$n = \frac{(t_{1-\alpha} + t_{1-\beta})^2 s^2}{\delta^2}$	Stat > Power and Sample Size > 1-sample t...
Two sample Z-test	Comparison of two unknown mean, σ known and equal	$n = \frac{2(Z_{1-\alpha} + Z_{1-\beta})^2 \sigma^2}{\delta^2}$	Stat > Power and Sample Size > 2-sample t...
Two sample t-test	Comparison of two unknown mean, σ 's unknown	$n = \frac{2(t_{1-\alpha} + t_{1-\beta})^2 s^2}{\delta^2}$	Stat > Power and Sample Size > 2-sample t...
Paired t-test	Two means of paired data with differences d_j	$n = \frac{(t_{1-\alpha} + t_{1-\beta})^2 s_d^2}{\delta^2}$	Stat > Power and Sample Size > Paired t...

These formulas are for one-sided tests. Replace α by $\alpha/2$ for a two sided test.



Testing Means Sample Size Example

Suppose you want to test if production mean has shifted from today to tomorrow. The difference you want to detect is 1.0 sigma of a measured variable. With alpha and beta risk of 5%, what sample size do you need?

Minitab: **Statistics > Power and Sample Size > 1 Sample t**

Python: `mqr.inference.mean.size_1sample`

Results

Difference	Sample Size	Target Power	Actual Power
1	16	0.95	0.961885

Sample Sizes for Other Parameters – Python

To determine the sample sizes for other parameters:

Test	Purpose	Equation	Python code
One sample χ^2 -test	Comparison to a reference standard deviation	$\chi_{1-\alpha,n}^2 = r^2 \chi_{\beta,n}^2$	<code>mqr.inference.stddev.size_1sample</code>
Two sample F-test	Comparison of two standard deviations	$F_{1-\alpha,n_1,n_2} = r^2 F_{\beta,n_1,n_2}$	<code>mqr.inference.stddev.size_2sample</code>
One Sample Rate test	Comparison of defect rate to a target	$n = \frac{\chi_{1-\beta,M}^2}{2R_A}, \chi_{\alpha,M}^2 = \left(\frac{R_A}{R_0}\right)^2 \chi_{1-\beta,M}^2$	<code>mqr.inference.rate.size_1sample</code>
Two Sample Rate test	Comparison of defect rate for two independent populations	$n = \frac{2(Z_\alpha \sqrt{R_1} + Z_\beta \sqrt{R_2})^2}{(R_1 - R_2)^2}$	<code>mqr.inference.rate.size_2sample</code>
One Sample Proportion test	Comparison of defective rate to a target	$n = \frac{(Z_\alpha + Z_\beta)^2}{4(\sin^{-1} \sqrt{\pi_0} - \sin^{-1} \sqrt{\pi_A})^2}$	<code>mqr.inference.proportion.size_1sample</code>
Two Sample Proportion test	Comparison of defective rate for two independent populations	$n = \frac{2(Z_\alpha + Z_\beta)^2}{4(\sin^{-1} \sqrt{\pi_1} - \sin^{-1} \sqrt{\pi_2})^2}$	<code>mqr.inference.proportion.size_2sample</code>

These formulas are for one-sided tests. Replace α by $\alpha/2$ for a two sided test.

Sample Sizes for Other Parameters – Minitab

To determine the sample sizes for other parameters:

Test	Purpose	Equation	Minitab
One sample χ^2 -test	Comparison to a reference standard deviation	$\chi_{1-\alpha,n}^2 = r^2 \chi_{\beta,n}^2$	Stat > Power and Sample Size > 1 Variance...
Two sample F-test	Comparison of two standard deviations	$F_{1-\alpha,n_1,n_2} = r^2 F_{\beta,n_1,n_2}$	Stat > Power and Sample Size > 2 Variances...
One Sample Rate test	Comparison of defect rate to a target	$n = \frac{\chi_{1-\beta,M}^2}{2R_A}, \chi_{\alpha,M}^2 = \left(\frac{R_A}{R_0}\right)^2 \chi_{1-\beta,M}^2$	Stat > Power and Sample Size > 1 Sample Poisson Rate...
Two Sample Rate test	Comparison of defect rate for two independent populations	$n = \frac{2(Z_\alpha \sqrt{R_1} + Z_\beta \sqrt{R_2})^2}{(R_1 - R_2)^2}$	Stat > Power and Sample Size > 2 Sample Poisson Rates...
One Sample Proportion test	Comparison of defective rate to a target	$n = \frac{(Z_\alpha + Z_\beta)^2}{4(\sin^{-1} \sqrt{\pi_0} - \sin^{-1} \sqrt{\pi_A})^2}$	Stat > Power and Sample Size > 1 Proportion...
Two Sample Proportion test	Comparison of defective rate for two independent populations	$n = \frac{2(Z_\alpha + Z_\beta)^2}{4(\sin^{-1} \sqrt{\pi_1} - \sin^{-1} \sqrt{\pi_2})^2}$	Stat > Power and Sample Size > 2 Proportions...

These formulas are for one-sided tests. Replace α by $\alpha/2$ for a two sided test.

Testing Standard Deviation Sample Size

When performing tests of standard deviations, the critical difference is best stated as a ratio r , rather than an offset δ .

- State the alternative hypothesis to reflect a percent difference that you don't want to miss in the form of a ratio.

$$r = \frac{\sigma_A}{\sigma_0}$$

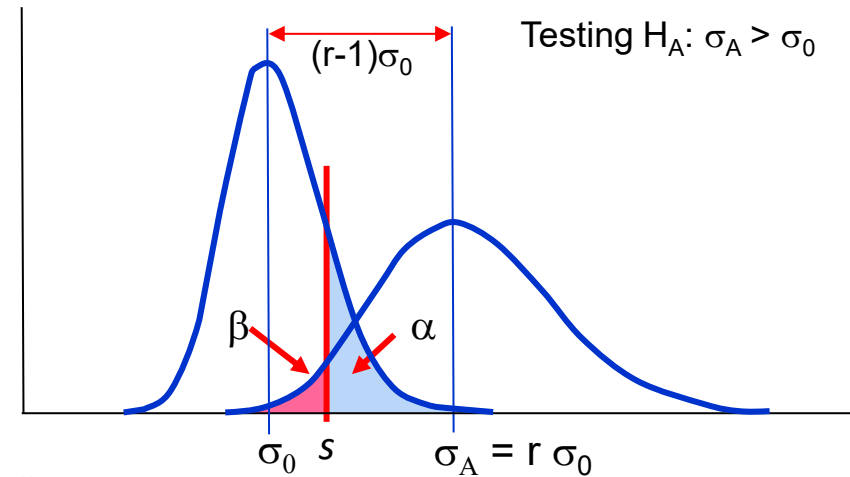
For example

- $H_0: \sigma = \sigma_0$
- $H_A: \sigma_A = 1.3\sigma_0$

"If σ is 30% larger we reject the null that they are the same."

Then the sample size equation is to find n such that

$$\chi^2_{1-\alpha, n} = r^2 \chi^2_{\beta, n}$$





Testing Standard Deviation Sample Size

Suppose you wish to determine if the variance of parts produced today is within 20% of a target. With $\alpha = \beta = 0.05$, how many units do you need to measure in your sample?

Minitab: **Statistics > Power and Sample Size > 1 Variance**

Python: `mqr.inference.stddev.size_1sample`

Results

Ratio	Sample Size	Target Power	Actual Power
0.833333	165	0.95	0.950400

Testing Difference of Two Standard Deviations

When performing tests of standard deviations, the critical difference is best stated as a ratio r , rather than a offset δ .

- State the alternative hypothesis to reflect a percent difference that you don't want to miss in the form of a ratio.

$$r = \frac{\sigma_2}{\sigma_1}$$

For example

- $H_0: \sigma_1 = \sigma_2$
- $H_A: \sigma_2 = 1.3\sigma_1$
"If σ_2 is 30% larger than σ_1 we reject the null that they are the same."

Then the sample size equation is to find n_1, n_2 such that

$$F_{1-\alpha, n_1, n_2} = r^2 F_{\beta, n_1, n_2}$$



Testing Two Standard Deviations Sample Size

Supplier A and B are potential part suppliers. If the two have equal variation, then Supplier B will be chosen for cost. However, if the variation of Supplier B is 20% larger than A's, you want to choose A. How many randomly sampled parts should you inspect?

Minitab: **Statistics > Power and Sample Size > 2 Variances**

Python: `mqr.inference.stddev.size_1sample`

Here $r = 1.2/1.0$ or $1.0/1.2$ (either, it does not matter):

Results

Ratio	Sample Size	Target Power	Actual Power
0.833333	328	0.95	0.950252

The sample size is for each group.

Testing Defect Rate Sample Size

A **rate** is often used as a unit of measure when more than one defect per unit “length” can occur (e.g. bad spots per sq. inch, number of scratches per cover, etc.)

When performing tests of rates, the critical difference will also be stated as a ratio. However, an estimate of the rate (corresponding to H_0) must also be included.

For example:

- $H_0: R_A = R_0$ Reject only α percent of time when true
- $H_A: R_A = 1.2R_0$ Reject only β percent of time when true

Then the sample size equation is to first find M such that

$$\chi_{\alpha, M}^2 = \left(\frac{R_A}{R_0} \right)^2 \chi_{1-\beta, M}^2, \text{ and then } n = \frac{\chi_{1-\beta, M}^2}{2R_A}$$

Defect Rate Sample Size Example

How many samples are needed to determine if the defect rate has changed from the historical rate of 2.5 defects per unit? You would like to know the defect rate has increased above 3.5 defects per unit. How many samples do you need?

Minitab: **Statistics > Power and Samples Size > 1 Sample Poisson Rate**

Python: `mqr.inference.rate.size_1sample`

Results

Comparison

Rate	Sample Size	Target Power	Actual Power
3.5	25	0.9	0.900159

Testing Two Rates Sample Size

When performing tests of rates, the critical difference will also be stated as a ratio. However, an estimate of the rate when the two rates are equal (corresponding to H_0) must also be included.

For example:

- $H_0: R_1 = R_2$ Reject only α percent of time when true
- $H_A: R_1 = 1.2R_2$ Reject only β percent of time when true

Then an approximate sample size equation is to find n such that

$$n = \frac{2(Z_\alpha\sqrt{R_1} + Z_\beta\sqrt{R_2})^2}{(R_1 - R_2)^2}$$

Testing Two Rates Sample Size

How many samples are needed to determine if the defect rate changes from one day to another day. The defect rate historically been 2.5 defects per unit. You would like to know the defect rate has increased above 3 defects per unit. How many samples do you need each day, if you are comparing one day to the next day?

Minitab: **Statistics > Power and Sample Size > 2 Sample Poisson Rate**

Python: `mqr.inference.rate.size_2sample`

Results

Comparison

Rate	Sample Size	Target Power	Actual Power
3	239	0.95	0.950646

The sample size is for each group.

Testing Proportions Sample Size

A **proportion** is often used as a unit of the fraction of defectives. For large enough sample sizes, a normal distribution approximation to the binomial distribution is possible.

For one sample and a target π_0 :

- $H_0: \pi = \pi_0$ Reject only α percent of time when true
- $H_A: \pi = \pi_A = \pi_0 + \delta$ Reject only β percent of time when true

For one sample and a target π_0 :

- $H_0: \pi_2 = \pi_1$ Reject only α percent of time when true
- $H_A: \pi_2 = \pi_1 + \delta$ Reject only β percent of time when true

Then the sample size equation is to first find M such that

$$n = \frac{(Z_\alpha + Z_\beta)^2}{4(\sin^{-1} \sqrt{\pi_0} - \sin^{-1} \sqrt{\pi_A})^2}$$



Testing Proportion Sample Size

How many samples are needed to determine if the defect rate changes from one day to another day. The defect rate historically been 20 defects per 100. You would like to know the defect rate has increased above 30 defects per 1000. For 90% confidence and power, how many samples do you need each day, if you are comparing one day to the next day?

Minitab: **Statistics > Power and Sample Size > 1 Proportion**

Python: `mqr.inference.proportion.size_1sample`

Results

Comparison p	Sample Size	Target Power	Actual Power
0.03	1585	0.9	0.900090

Testing Means Sample Size Example – Python

Suppose you want to test if production mean has shifted from today to tomorrow. The difference you want to detect is 1.0 sigma of a measured variable. With alpha and beta risk of 5%, what sample size do you need?

```
from statsmodels.stats.power import tt_solve_power

beta = 0.05      # 1 - Power
alpha = 0.05     # Significance level
effect = 1.0     # effect = delta/sigma
alt = 'two-sided' # 'smaller', 'two-sided', 'larger'

result = tt_solve_power(effect_size=effect,
                        power=1.0-beta, nobs = None,
                        alternative=alt, alpha=alpha)

result = int(round(result))
print('Sample size:', result)
```

Sample size: 15

Testing Means Sample Size Example – Minitab

Suppose you want to test if production mean has shifted from today to tomorrow. The difference you want to detect is 1.0 sigma of a measured variable. With alpha and beta risk of 5%, what sample size do you need?

Power and Sample Size for 1-Sample t

Specify values for any two of the following:

Sample sizes:

Differences:

Power values:

Standard deviation:

Options... Graph...

Help OK Cancel

Results

Difference	Sample Size	Target Power	Actual Power
1	16	0.95	0.961885

Testing Standard Deviation Sample Size – Python

Suppose you wish to determine if the variance of parts produced today is within 20% of a target. With $\alpha = \beta = 0.05$, how many units do you need to measure in your sample?

```
from scipy.stats import chi2
import numpy as np

chi2_sample_size(1.2, 0.05, 0.05, "<")
```

164

```
def chi2_sample_size(r, alpha, beta, test):
    n = 1
    if test == "<":
        NP1 = alpha
        DP1 = 1-beta
    elif test == ">":
        NP1 = 1-beta
        DP1 = 1-alpha
    else:
        NP1 = beta
        DP1 = (1-alpha/2)
    while n < 1000:
        num = chi2.ppf(1-NP1, df=n)
        denom = chi2.ppf(1-DP1, df=n)
        if test == ">":
            rat = 1.0/np.sqrt(num/denom)
        else:
            rat = np.sqrt(num/denom)
        if rat < r:
            break
        n+= 1
    if n == 1000: raise ValueError("n > 1000.")
    return n
```

Testing Standard Deviation Sample Size – Minitab

Suppose you wish to determine if the variance of parts produced today is within 20% of a target. With $\alpha = \beta = 0.05$, how many units do you need to measure in your sample?

Power and Sample Size for 1 Variance

Enter ratios of standard deviations ▼

Specify values for any two of the following:

Sample sizes:

Ratios:

(StDev / hypothesized StDev)

Power values:

Options... Graph...

Help OK Cancel

Power and Sample Size for 1 Variance: Options

Alternative Hypothesis

☒ Less Than

☐ Not equal

☐ Greater Than

Significance level:

Help OK Cancel

Results			
Ratio	Sample Size	Target Power	Actual Power
0.833333	165	0.95	0.950400

Two σ Test Sample Size Example – Python

Supplier A and B are potential part suppliers. If the two have equal variation, then Supplier A will be chosen for cost. However, if the variation of Supplier B is 20% lower than A's, you want to choose B. How many randomly sampled parts should you inspect?

```
from scipy.stats import f as fdist
import numpy as np

fctest_sample_size(1.2, 0.05, 0.05, ">")
```

328

```
from scipy.stats import f as fdist
import numpy as np

def fctest_sample_size(r, alpha, beta, test):
    n = 1
    if test == "<":
        if r > 1.0:
            print('diff must be > 1 for a greater-than alternative
hypothesis.')
            return "Nan"
        r = 1.0/r
        NP1 = 1-alpha
        DP1 = beta
    elif test == ">":
        if r < 1.0:
            print('diff must be < 1 for a less-than alternative hypothesis.')
            return "Nan"
        NP1 = 1-alpha
        DP1 = beta
    else:
        NP1 = (1-alpha/2)
        DP1 = beta
        if r < 1.0: r = 1.0/r

    while n < 10000:
        num = fdist.ppf(NP1, n, n)
        denom = fdist.ppf(DP1, n, n)
        rat = np.sqrt(num/denom)
        if rat < r:
            break
        n+= 1
    if n == 10000: raise ValueError("n > 10000.")
    return n+1
```

Two σ Test Sample Size Example – Minitab

Supplier A and B are potential part suppliers. If the two have equal variation, then Supplier A will be chosen for cost. However, if the variation of Supplier B is 20% lower than A's, you want to choose B. How many randomly sampled parts should you inspect?

Power and Sample Size for 2 Variances

Enter ratios of standard deviations

Specify values for any two of the following:

Sample sizes:

Ratios:

(StDev 1 / StDev 2)

Power values:

Options... Graph... Help OK Cancel

Power and Sample Size for 2 Variances: Options

Alternative Hypothesis

☒ Less Than

☐ Not equal

☐ Greater Than

Significance level:

Method

☐ Levene's Test

☒ F Test

Help OK Cancel

Results

Ratio	Sample Size	Target Power	Actual
0.833333	328	0.95	0.950252

The sample size is for each group.

Rate Sample Size Example – Python

How many samples are needed to determine if the defect rate has changed from the historical rate of 2.5 defects per unit? You would like to know the defect rate has increased above 3.5 defects per unit. How many samples do you need?

```
from scipy.stats import f as fdist
import numpy as np

onerate_sample_size(3.5, 2.5, 0.05, 0.1, "<")
```

25

```
def onerate_sample_size(ra, r0, alpha, beta, test):
    n = 1
    r = ra/r0
    if test == "<":
        NP1 = 1-beta
        DP1 = alpha
    elif test == ">":
        NP1 = 1-beta
        DP1 = alpha
    else:
        NP1 = 1-beta
        DP1 = alpha/2
    while n < 10000:
        num = chi2.ppf(NP1, df=n)
        denom = chi2.ppf(DP1, df=n)
        rat = num/denom
        if rat < r:
            break
        n+= 1
    if n == 10000: raise ValueError("n > 10000.")
    return int(round(num/2.0/np.maximum(ra,r0)))
```

Rate Sample Size Example – Minitab

How many samples are needed to determine if the defect rate has changed from the historical rate of 2.5 defects per unit? You would like to know the defect rate has increased above 3.5 defects per unit. How many samples do you need?

Power and Sample Size for 1-Sample Poisson Rate

Specify values for any two of the following:

Sample sizes:

Comparison rates:

Power values:

Hypothesized rate:

Options... Graph... Help OK Cancel

Power and Sample Size for 1-Sample Poisson Rate: Options

Alternative Hypothesis

☐ Less Than

☐ Not equal

☒ Greater Than

Significance level:

"Length" of observation (time, items, area, volume, etc.):

Help OK Cancel

Results

Comparison

Rate	Sample Size	Target Power	Actual Power
3.5	33	0.95	0.953564

Testing Two Rates Sample Size – Python

How many samples are needed to determine if the defect rate changes from one day to another day. The defect rate historically been 2.5 defects per unit. You would like to know the defect rate has increased above 3 defects per unit. How many samples do you need each day, if you are comparing one day to the next day?

```
from scipy.stats import norm
import numpy as np

def tworate_sample_size(ra, r0, alpha, beta, test):
    if test == "!=": alpha = alpha*0.5
    Zb = norm.ppf(1-beta, loc=0.0, scale=1.0)
    Za = -1.0*norm.ppf(alpha, loc=0.0, scale=1.0)
    num = Za*np.sqrt(r0)+Zb*np.sqrt(ra)
    nn = 2.0*(num*num)/(ra-r0)/(ra-r0)
    return int(round(nn))
```

```
ra = 3.0          # Alternative rate at which to reject
r0 = 2.5          # Nominal rate to test against
alpha = 0.05      # significance
beta = 0.05       # 1 - power
alt = "!="        # '<', '>', '!=' Alternative Hypothesis

result = tworate_sample_size(ra, r0, alpha, beta, alt)
print('Sample size needed for each sample =', result)
```

Sample size needed for each sample = 238

Testing Two Rates Sample Size – Minitab

How many samples are needed to determine if the defect rate changes from one day to another day. The defect rate historically been 2.5 defects per unit. You would like to know the defect rate has increased above 3 defects per unit. How many samples do you need each day, if you are comparing one day to the next day?

Power and Sample Size for 2-Sample Poisson Rate

Specify values for any two of the following:

Sample sizes:

Comparison rates (R1):

Power values:

Baseline rate (R2):

Options... Graph...

Help OK Cancel

Power and Sample Size for 2-Sample Poisson Rate: Options

Alternative Hypothesis

☐ Less than ($R1 < R2$)

☐ Not equal ($R1 \neq R2$)

☒ Greater than ($R1 > R2$)

Significance level:

"Lengths" of observation (time, items, area, volume, etc.):

Help OK Cancel

Results

Comparison

Rate	Sample Size	Target Power	Actual Power
3	239	0.95	0.950646

The sample size is for each group.

Testing Proportion Sample Size – Python

How many samples are needed to determine if the defective rate is below target. The defect rate historically been 20 defects per 100. You would like to know the defect rate has increased to above 30 defects per 1000. For 90% confidence and power, how many samples do you need each day?

```
def prop_sample_size(p1, p2, alpha, beta, test):  
  
    if test == "!=": alpha = alpha*0.5  
  
    Zb = norm.ppf(1-beta,loc=0.0,scale=1.0)  
    Za = -1.0*norm.ppf(alpha,loc=0.0,scale=1.0)  
    num = (Za+Zb)  
    denom = np.arcsin(np.sqrt(p1)) -  
            np.arcsin(np.sqrt(p2))  
    denom = denom*denom  
    nn = 0.25*(num*num)/denom  
  
    return int(round(nn))
```

```
from scipy.stats import norm  
import numpy as np  
  
prop_sample_size(20.0/1000.0, 30.0/1000.0,  
                 0.10, 0.10, "<")
```

Testing Proportion Sample Size – Minitab

How many samples are needed to determine if the defective rate is below target. The defect rate historically been 20 defects per 100. You would like to know the defect rate has increased to above 30 defects per 1000. For 90% confidence and power, how many samples do you need each day?

Power and Sample Size for 1 Proportion

Specify values for any two of the following:

Sample sizes:

Comparison proportions:

Power values:

Hypothesized proportion:

Options... Graph...

Help OK Cancel

Power and Sample Size for 1 Proportion: Options

Alternative Hypothesis

☐ Less than

☐ Not equal

☒ Greater than

Significance level:

Help OK Cancel

Results

Comparison p	Sample Size	Target Power	Actual
0.03	1585	0.9	0.900090

Testing Two Proportions Sample Size – Python

How many samples are needed to determine if the defective has changed from this week to next week. The historical defect rate is 200 defects per 1000. You would like to know the defective rate has increased above 300 defects per 1000. For 90% confidence and power, how many samples do you need?

```
import numpy as np

prop_sample_size(200.0/1000.0,
                 300.0/1000.0, 0.10, 0.10, "<")
```

```
def twoprop_sample_size(p1, p2, alpha, beta, test):

    if test == "!=": alpha = alpha*0.5

    Zb = norm.ppf(1-beta,loc=0.0,scale=1.0)
    Za = -1.0*norm.ppf(alpha,loc=0.0,scale=1.0)
    num = (Za+Zb)
    denom = np.arcsin(np.sqrt(p1)) -
            np.arcsin(np.sqrt(p2))
    denom = denom*denom
    nn = 0.25*(num*num)/denom

    return 2*int(round(nn))
```

Testing Two Proportions Sample Size – Minitab

How many samples are needed to determine if the defective has changed from this week to next week. The historical defect rate is 200 defects per 1000. You would like to know the defective rate has increased above 300 defects per 1000. For 90% confidence and power, how many samples do you need?

Power and Sample Size for 2 Proportions

Specify values for any two of the following:

Sample sizes:

Comparison proportions (p1):

Power values:

Baseline proportion (p2):

Options... Graph...

Help OK Cancel

Power and Sample Size for 2 Proportions: Options

Alternative Hypothesis

☐ Less than ($p_1 < p_2$)

☐ Not equal ($p_1 \neq p_2$)

☒ Greater than ($p_1 > p_2$)

Significance level:

Help OK Cancel

Results

Comparison p	Sample Size	Target Power	Actual Power
0.03	245	0.9	0.900266
The sample size is for each group.			



Overview

Test Errors

Statistical Inferencing

T-test

Parametric tests

Sample Sizes

Non-parametric tests

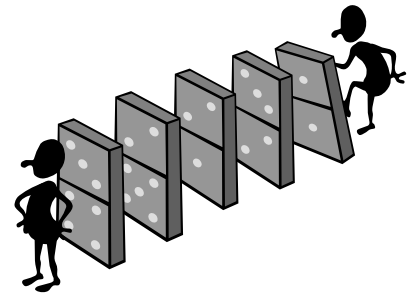
What are Non-Parametric Statistics?

Non-Parametric Statistics, or Distribution-Free Statistics, is the branch of statistics that deal with data that:

- Does not utilize the estimation of variance, mean, and any other population parameters
- Does not make any assumption about the distribution of the sampled populations

It computes results based upon the rank order of the data points – 1st, 2nd, 3rd, ...

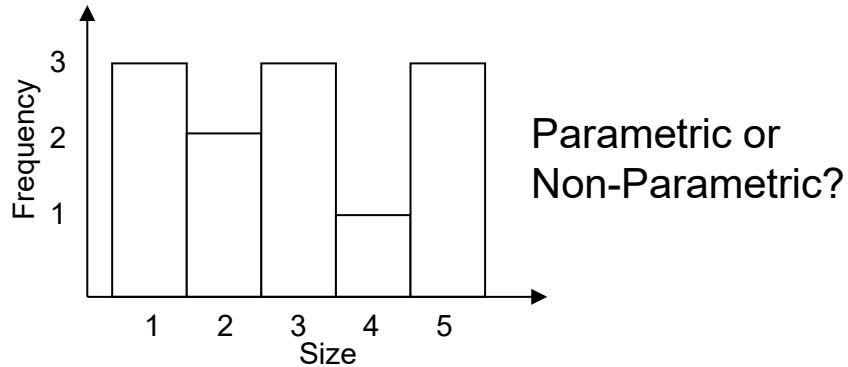
- Similar to a median or quartile range calculation



Applications of Non-Parametric Tests

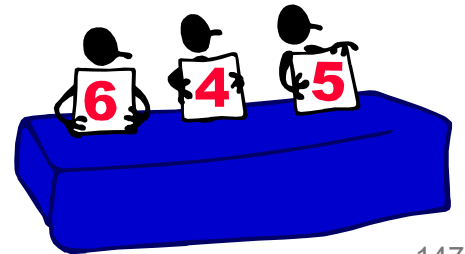
When the data cannot be modeled by a distribution, like non-normally distributed

When the sample size for the data set is small



When the data is ranked (ordinal scale)

- Severe, Moderate, Light, None
- Upper, Upper Middle, Middle, Lower Middle, Lower



Disadvantages of Non-Parametric Tests

Less powerful / efficient than the corresponding Parametric tests

Use the parametric tests if the assumptions of are met

- Normality
- Constant variances
- Existence of a distribution

Otherwise you cannot and must use the less powerful non-parametric tests



Non-Parametric Hypothesis Tests

Nonparametric Test	Test For	Parametric equivalent	# of Factors	# of Levels
Run test	Randomness	Distribution Fit Tests (Normality, ...)	1	
1-Sample Sign	Central Location	1 sample t Paired t	1	1
1-Sample Wilcoxon			1	2
Man-Whitney	Equality of Two Population Locations	2 sample t	1	2
Kruskal-Wallis	Equality of Multiple Population Locations	ANOVA	1	2+
Moods Median			1	2+
Friedman	Randomized Block Experiment	2 way ANOVA	2	2+
Levene's	Dispersion	Chi-2 test	1	1
		F-test Bartlett's test	1	2+



Runs Test: Is the Data Random?

A **runs test** is used to determine if the element of a sample are mutually independent. If the data fails a runs test, then even the non-parametric tests cannot be used.

The sample data is converted into +’s and –’s, such as comparing to the median. Then length of each run of +’s and –’s is counted.

For example + + + + – – – + + + – – + + + + + – – – – has 22 elements and $R = 6$ runs. There are 3 positive runs, $R^+ = 3$, and 3 negative runs, $R^- = 3$.

Runs Test: Is the Data Random?

A **runs test** is used to determine if the element of a sample are mutually independent.

For example $++++--++--++++--$ has 22 elements and $R = 6$ runs.

There are 3 positive runs, $R^+ = 3$, and 3 negative runs, $R^- = 3$.

The number of runs is a normally distributed random variable with

$$\bar{R} = \frac{2R^+R^-}{R} + 1 \quad \text{and} \quad s^2 = \frac{(\mu-1)(\mu-2)}{R-1}$$

Then a runs test statistic is $Z = \frac{R - \bar{R}}{s}$

and $p = 2\min(\Phi^{-1}(Z), 1 - \Phi^{-1}(Z))$

Runs Test Example

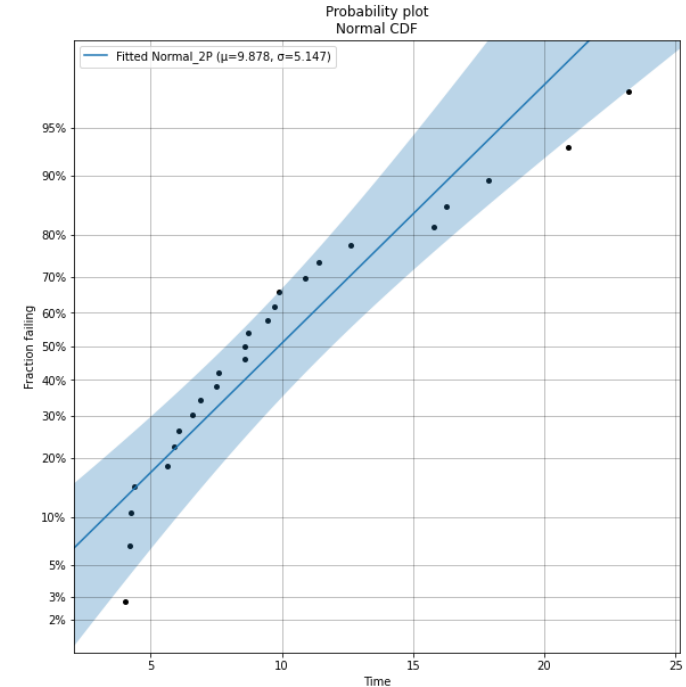
A manufacturer is producing engines whose rated life ought be 10 years. A random sample of 25 units was taken to see how long each lasted before failure and repair. The life data is as shown. Is the life centered around 10 years?

12.62	9.45	06.05
9.71	23.23	10.90
8.59	7.60	06.89
11.42	4.23	08.72
5.90	6.59	07.52
17.86	4.38	05.64
9.90	8.60	04.22
16.25	20.91	04.01
15.77		

Runs Test Example

Perhaps a t-test against a mean life of 10 years.

Can we do a t-test? The data must be normal. Is it?



Anderson Darling AD=0.995
p-value=0.011

P is low. Data is not normal.
We cannot use the t-test.

Runs Test Example – Python

Perhaps a sign test against a median life of 10 years.

Can we do a sign test? The data must be random. Is it?

To test for randomness, we use a runs test.

```
co = 'median' # 'mean', 'median' the cutoff midpoint for the runs test
mgr.inference.nonparametric.dist.test_1sample(data, cutoff=co)
```

Hypothesis Test	
randomness	
method	runs
H_0	<code>dist(x) == iid</code>
H_1	<code>dist(x) != iid</code>
statistic	-0.401101
p-value	0.688346

p is not low. Data is random.
We can use non-parametric tests.

Runs Test Example – Minitab

Perhaps a sign test against a median life of 10 years.

Can we do a sign test? The data must be random. Is it?
To test for randomness, we use a runs test.

Test

Null hypothesis H_0 : The order of the data is random

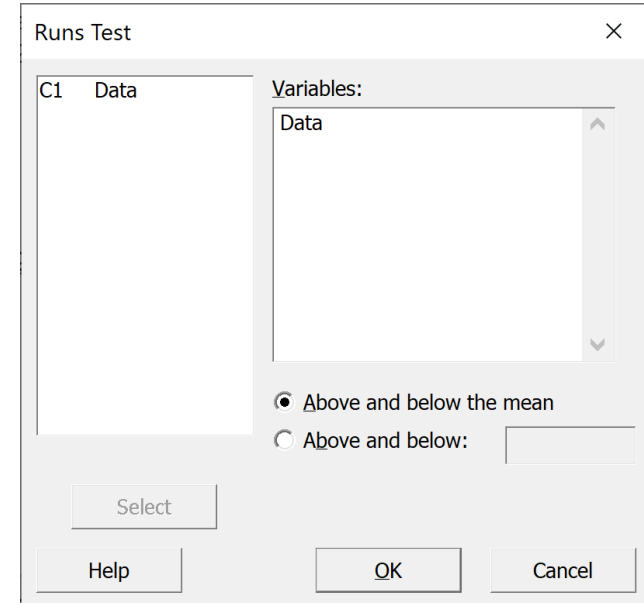
Alternative hypothesis H_1 : The order of the data is not random

Number of Runs

Observed	Expected	P-Value
12	12.52	0.817

The p-value may not be accurate for samples with fewer than 11 observations above K or fewer than 11 below.

p is not low. Data is random.
We can use non-parametric tests.



Runs Test Example – Python

An example of non-random data is collected by H. S. Lew of NIST in 1969 to measure steel-concrete beam deflections as time-series data. It is time-correlated and not random.

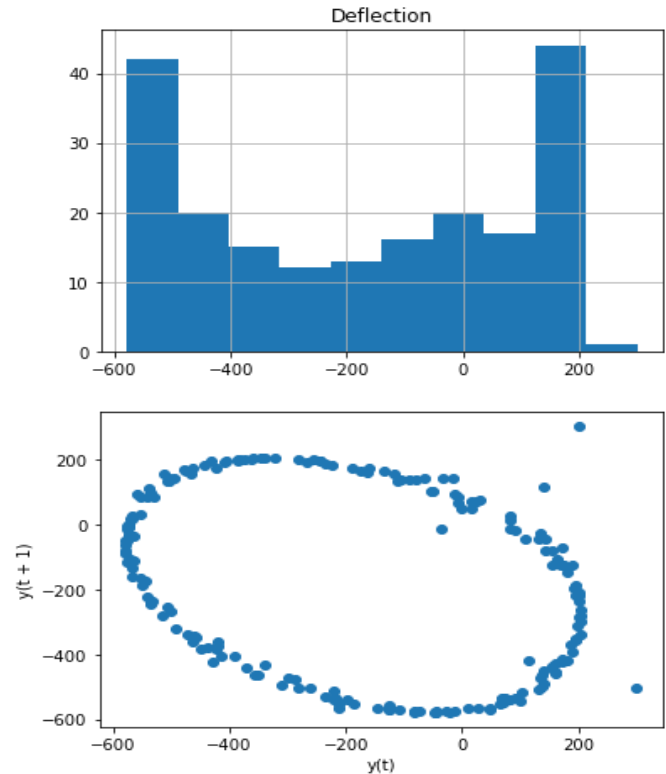
```
import pandas as pd
from statsmodels.sandbox.stats.runs import runstest_1samp

url = 'https://www.itl.nist.gov/div898/handbook/datasets/LEW.DAT'
df = pd.read_csv(url, skiprows=24, header=0,
                 names=['Deflection'], dtype=np.float64)

mgr.inference.nonparametric.dist.test_1sample(
    df['Deflection'], cutoff='mean', method='runs')

with Figure(4, 3) as (fig, ax):
    sns.histplot(df, ax=ax)

with Figure(5, 4) as (fig, ax):
    pd.plotting.lag_plot(df, alpha=0.5, ax=ax)
```



p is low. Data is not random.
We cannot even use non-parametric tests.



Tests for Location (Central Tendency)

Do the populations have similar medians?

Sign Test

Wilcoxon Signed-Rank Test

Mann-Whitney Test

Kruskal-Wallis Test

Mood's Median Test

Friedman Test

Sign Test and the Wilcoxon Signed-Rank Test

May be used for:

Verifying if a population's median is equal to a target value η_0

- $H_0: \eta_X = \eta_0$ where η_0 is a target value
- $H_{A1}: \eta_X < 0$
- $H_{A2}: \eta_X > 0$
- $H_{A3}: \eta_X \neq 0$

Paired comparison of 2 populations

- $H_0: \eta_D = 0$ where η_D is the median of $X-Y$
- $H_{A1}: \eta_X < 0$
- $H_{A2}: \eta_X > 0$
- $H_{A3}: \eta_X \neq 0$

Sign Test

Interested in only the direction (+/-), rather than the quantitative amount, of the difference of the variable from the proposed median

Procedure

1. For a sample (X_i and a target η_0) or for a matched pair of observations (X_i and Y_i), determine the difference of $D_i = X_i - \eta_0$ or $D_i = X_i - Y_i$.
2. Check to ensure data is random using the run test
3. Determine the “sign” (+, -) for each D_i . “0s” are deleted. Let $m = n - \text{number of “0s”}$.
4. For $H_A: \eta_D < 0$: Determine $R = R^+$ = the number of positive D_i 's.
For $H_A: \eta_D > 0$: Determine $R = R^-$ = the number of negative D_i 's.
For $H_A: \eta_D \neq 0$: Determine both R^+ and R^- . Use $R = \min(R^+, R^-)$
5. Compute a p-value for R using a binomial distribution of m trials, each with probability 0.5. For $H_A: \eta_D \neq 0$ multiply the binomial result by 2.
6. Reject $H_0: \eta_D = 0$ if $p < \alpha$

Example 1: Sign Test

An experiment was conducted to study the effectiveness of a new drug.

- 10 patients were chosen at random.
- They rated their condition according to the severity (1=least severe, 5=most severe).
- After taking the drug, the patients rated their condition again.

We would like to know if the drug is effective in reducing the perceived severity.

$$H_0: \eta_D = 0$$

$$H_A: \eta_D < 0$$

Severity Before	Severity After
2	2
2	1
1	2
1	1
3	1
4	2
5	3
3	2
1	1
5	1

Example 1: What the Sign Test Does

The Sign Test is actually a binomial test, assuming that the grouping is supposed to be split 50-50.

Compute the sign test statistic

$$R = R^+ = \# \text{ + 'ves} = 1$$

Severity Before	Severity After	Delta	Sign
2	2	0	0
2	1	-1	-1
1	2	1	+1
1	1	0	0
3	1	-2	-1
4	2	-2	-1
5	3	-2	-1
3	2	-1	-1
1	1	0	0
5	1	-4	-1

Example 1: Compute P-Value

What are the chances that we'd get one “+” out of seven if they are supposed to happen evenly?

- Use the binomial distribution with $R = 1$, $m = 10 - 3 = 7$, $p = 0.5$:

$$p - \text{value} = \sum_{r=0}^R \frac{m!}{r! (m-r)!} 0.5^r 0.5^{n-r}$$

$$p - \text{value} = \sum_{r=0}^1 \frac{7!}{r! (7-r)!} 0.5^r 0.5^{n-r}$$

$$p - \text{value} = \frac{7!}{0! 7!} 0.5^0 0.5^7 + \frac{7!}{1! 6!} 0.5^1 0.5^6 = 0.0625$$

“Accept” or reject H_0 : “ $\eta_D = 0$ ”?

- $p = 0.0625 > \alpha = 0.05$, so we cannot reject H_0
- We cannot reject $\eta_D = 0$ with 95% confidence.

We cannot reject that the “after” median is the same as the “before” median.

We ‘accept’ that medians are the same. The drug didn’t improve anything.

Example 1: Sign Test – Python

```
sample1 = np.array([2.0, 2.0, 1.0, 1.0, 3.0, 4.0, 5.0, 3.0, 1.0, 5.0])
sample2 = np.array([2.0, 1.0, 2.0, 1.0, 1.0, 2.0, 3.0, 2.0, 1.0, 1.0])
df = pd.DataFrame(data={'sample1': sample1, 'sample2': sample2})
```

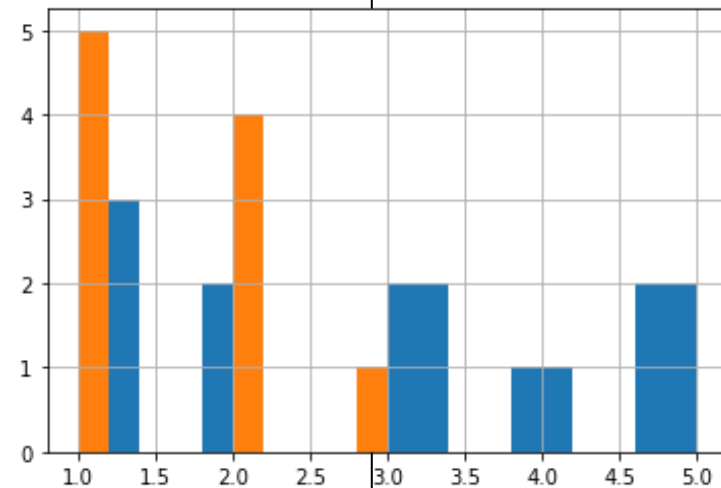
```
with Figure(5, 4) as (fig, ax):
    sns.histplot(df, multiple='dodge', ax=ax)
    plot = mqr.nbtools.grab_figure(fig)
```

```
# Display the medians alongside the plot and hypothesis test below.
# The stacking functions will render markdown, including a table.
medians_markdown = f'''
```

```
### Medians
```

```
| | |
|---| ---:|
| sample1 | **{np.median(sample1)** |
| sample2 | **{np.median(sample2)** |
'''
```

```
hstack(
    plot,
    hstack(
        vstack(
            mqr.inference.nonparametric.median.test_1sample(sample1-sample2, method='sign'),
            mqr.inference.nonparametric.median.test_nsample(sample1, sample2)),
        medians_markdown
    )
)
```

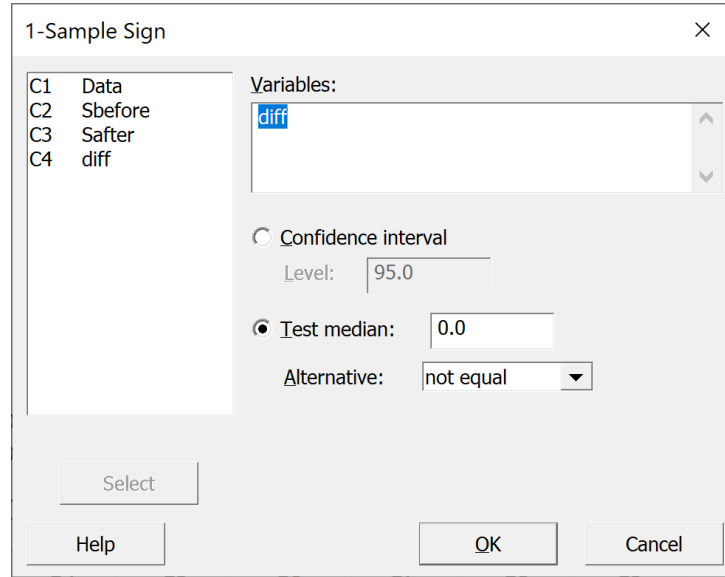


H0: medians are the same: $p = 0.0625$

We 'accept' that medians are the same.
The drug didn't improve anything.

Example 1: Sign Test – Minitab

Using Minitab



1-Sample Sign

Variables:

diff

☐ Confidence interval

Level: 95.0

☒ Test median: 0.0

Alternative: not equal

Select

Help OK Cancel

Test

Null hypothesis

$H_0: \eta = 0$

Alternative hypothesis $H_1: \eta \neq 0$

Sample	Number < 0	Number = 0	Number > 0	P-Value
diff	6	3	1	0.125

We 'accept' that medians are the same.
The drug didn't change anything.



Wilcoxon Signed-Rank Test

Utilizes not only the direction (Sign Test), but also the magnitude of the difference

A more powerful test

Requires the distribution be symmetric and continuous

- Mean equals the median

If the assumptions of the parametric t-test are met, then the Wilcoxon test is approximately 95% efficient.

Wilcoxon Signed-Rank Test

Procedure

For each matched pair of observations (X_i and Y_i), determine the signed difference of $D_i = X_i - Y_i$ between the two variables.

Check for randomness with the Run Test

Sort the D_i based on amplitude in absolute value, and assign a rank order R_i .

Determine $W = \sum_{i=1}^n \text{sgn}(D_i)R_i$ where $\text{sgn}()$ is the sign function and R_i is the rank order

Determine the Z statistics as
$$Z_W = \frac{W}{\sqrt{\frac{n(n+1)(2n+1)}{6}}}$$

Determine the p-value of the test statistic as $p = \Pr(Z_{W^+})$ using the normal distribution

Reject $H_0: \eta_D = 0$ if $p < \alpha$

Example 1: Wilcoxon Test – Python

Revisit the previous example, but now consider the difference before and after.

```
data1 = np.array([
    0.0, -1.0, 1.0, 0.0, -2.0, -2.0, -2.0, -1.0, 0.0, -4.0
])
hyp = -1.0 # the hypothesized median.
alt = 'less' # 'less', 'two-sided', 'greater'

with Figure(4, 3) as (fig, ax):
    sns.histplot(data1, ax=ax)
    plot = mqr.nbtools.grab_figure(fig)

hstack(
    plot,
    mqr.inference.nonparametric.median.test_1sample(x=data1,
    H0_median=hyp, alternative=alt, method='wilcoxon'))
```

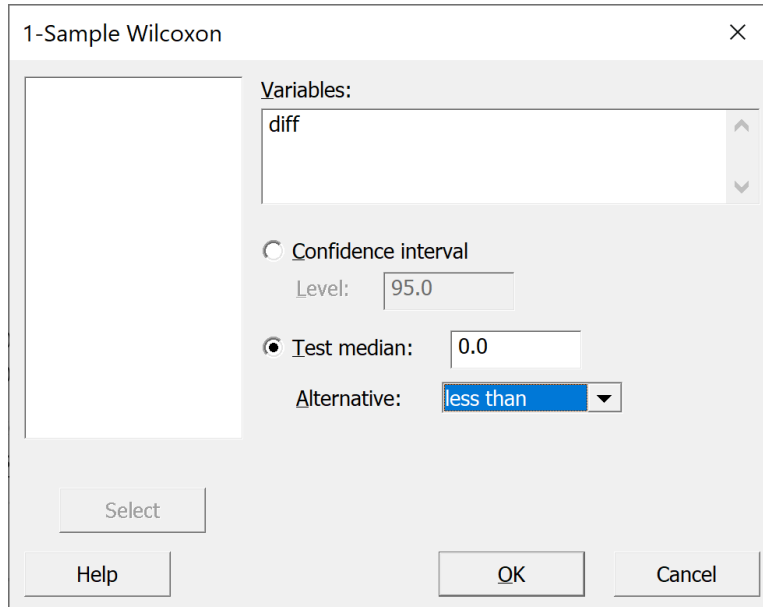
Now we can reject that medians are the same!
The drug did make a difference.
Note this is different from the sign test, the Wilcoxon test has more resolving power.

Severity Before	Severity After	Delta
2	2	0
2	1	-1
1	2	1
1	1	0
3	1	-2
4	2	-2
5	3	-2
3	2	-1
1	1	0
5	1	-4

Hypothesis Test	
median	
method	wilcoxon
H_0	median(x) == -1
H_1	median(x) < -1
statistic	17.5
p-value	0.470813

Example 1: Wilcoxon Test – Minitab

Revisit the previous example, but now consider the difference before and after.



The dialog box is titled "1-Sample Wilcoxon". It contains a "Variables:" list with "diff" entered. Below this, there are two radio buttons: "Confidence interval" (unselected) and "Test median:" (selected). The "Level:" for the confidence interval is set to "95.0". The "Test median:" is set to "0.0". Below that, the "Alternative:" dropdown menu is set to "less than". At the bottom, there are buttons for "Select", "Help", "OK", and "Cancel".

Test

Null hypothesis $H_0: \eta = 0$

Alternative hypothesis $H_1: \eta \neq 0$

Sample	N for Test	Wilcoxon Statistic	P-Value
diff	7	2.00	0.026

Now we can reject that medians are the same!
The drug did make a difference.

Note this is different from the sign test, the Wilcoxon test has more resolving power.



Mann-Whitney Test

For comparing the medians of two populations

- $H_0: \eta_X = \eta_Y$
- $H_{A1}: \eta_X < \eta_Y$
- $H_{A2}: \eta_X > \eta_Y$
- $H_{A3}: \eta_X \neq \eta_Y$

Calculates based on the direction and magnitude of the differences

Requires that two populations that have the same shape and equal variances



Mann-Whitney Test

Procedure

For two sets of unpaired observations (X_i and Y_i)

Check for randomness with the Run Test

Check for equal variances using the F-test

Rank the two combined samples, with the smallest observation given rank 1, the second smallest, rank 2, etc.
If two or more observations are tied, the average rank is assigned to each.

Calculate the test statistic R_1 as the sum of the ranks of the first sample.

$$\text{Then } U_1 = R_1 - \frac{n(n+1)}{2}$$

Determine the p-value of the U_1 test statistic

- For $n > 20$, calculate $p = \Pr(Z_U)$ using the normal distribution, where $Z_U = \frac{U_1 - \frac{n_1 n_2}{2}}{\sqrt{\frac{n_1 n_2 (n_1 + n_2 + 1)}{12}}}$

Reject H_0 if p is less than α

Example 2: Mann-Whitney Test – Python

An experiment was conducted to study the effectiveness of web-based instruction as compared to a live classroom attendance.

- Students are assigned at random to attend live or via virtual webmeeting call
- The first group of students will attend the lectures in the classroom
- The second group will attend via a webmeeting app

The students were then graded on learning outcomes.

Is there a difference between the grading outcomes?

Web	Live
74.4	85.9
82.6	84.4
80.9	83.4
79.4	74.0
78.4	81.9
69.0	84.8
76.9	85.5
79.8	78.9
80.5	78.4
73.9	77.2
73.4	79.4
72.2	87.6

Example 2: Mann-Whitney Test – Python

Run a Mann-Whitney 2 sample test.

```
import numpy as np
from scipy.stats import mannwhitneyu

#import pingouin as pg

web = np.array([74.4, 82.6, 80.9, 79.4, 78.4, 69.0,
                76.9, 79.8, 80.5, 73.9, 73.4, 72.2])
live = np.array([85.9, 84.4, 83.4, 74.0, 81.9, 84.8,
                85.5, 78.9, 78.4, 77.2, 79.4, 87.6])

mqr.inference.nonparametric.median.test_nsample(live,
web, alternative=alt, method='mann-whitney')
```

Hypothesis Test
equality of medians

method	mann-whitney
--------	--------------

H₀	median(x) - median(y) == 0
----------------------	----------------------------

H₁	median(x) - median(y) != 0
----------------------	----------------------------

statistic	4798
------------------	------

p-value	0.622563
----------------	----------

p is less than α .

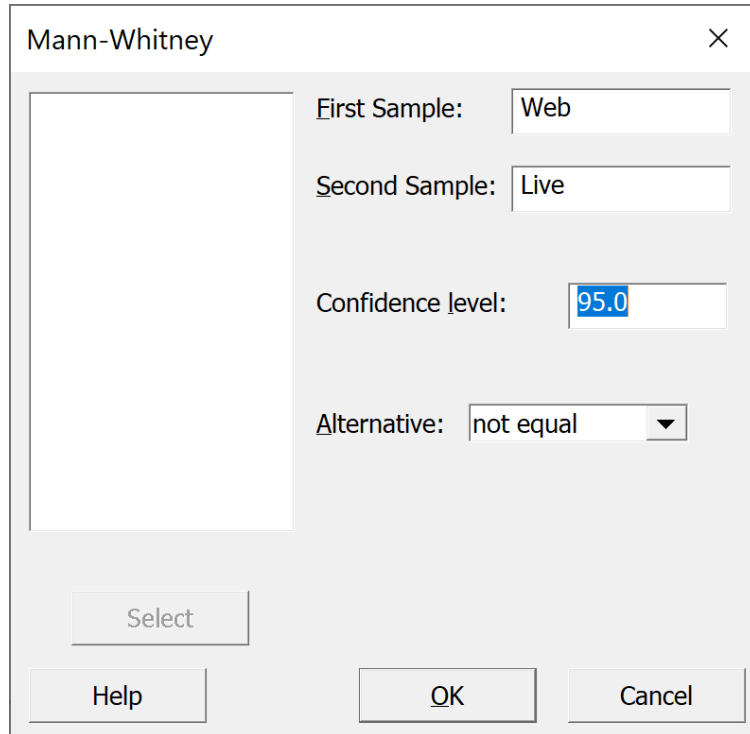
We reject H_0

We reject $\eta_1 = \eta_2$.

Live is better than web as taught.

Example 2: Mann-Whitney Test – Minitab

Run a Mann-Whitney 2 sample test.



Mann-Whitney

First Sample: Web

Second Sample: Live

Confidence level: 95.0

Alternative: not equal

Select

Help OK Cancel

Test

Null hypothesis $H_0: \eta_1 - \eta_2 = 0$

Alternative hypothesis $H_1: \eta_1 - \eta_2 \neq 0$

Method	W-Value	P-Value
Not adjusted for ties	108.00	0.017
Adjusted for ties	108.00	0.017

p is less than α .

We reject H_0

We reject $\eta_1 = \eta_2$.

Live is better than web as taught.



Kruskal-Wallis Test

Used for verifying if independent populations have a common central tendency

- $H_0: \eta_1 = \dots = \eta_k$ where η_i is the median of population i
- $H_A: \eta_i \neq \eta_j$ for some $i \neq j$



Kruskal-Wallis Test

Procedure

Rank all the observations of the k groups collectively, in ascending order.

Determine the values of $\bar{R}_i$ (the mean rank in group i) and $\bar{\bar{R}}$ (the overall mean rank)

Compute the test statistic $H = \frac{12}{N(N+1)} \sum_{i=1}^k n_i (\bar{R}_i - \bar{\bar{R}})^2$

Compute the p-value for H using a χ^2 -distribution with $k-1$ degrees of freedom.

Example 3: Kruskal-Wallis Test – Python

A survey at a local primary school was conducted to analyze the relationship between educational levels of the parents and number of times they attend their children's school activities.

High School	College	Graduate School
4	6	5
2	3	2
3	4	6
6	5	1
8	2	8
2		

What are useful null and alternate hypotheses?

$$H_0: \eta_1 = \eta_2 = \eta_3$$

$$H_A: \eta_i \neq \eta_j \text{ for one } i, j$$

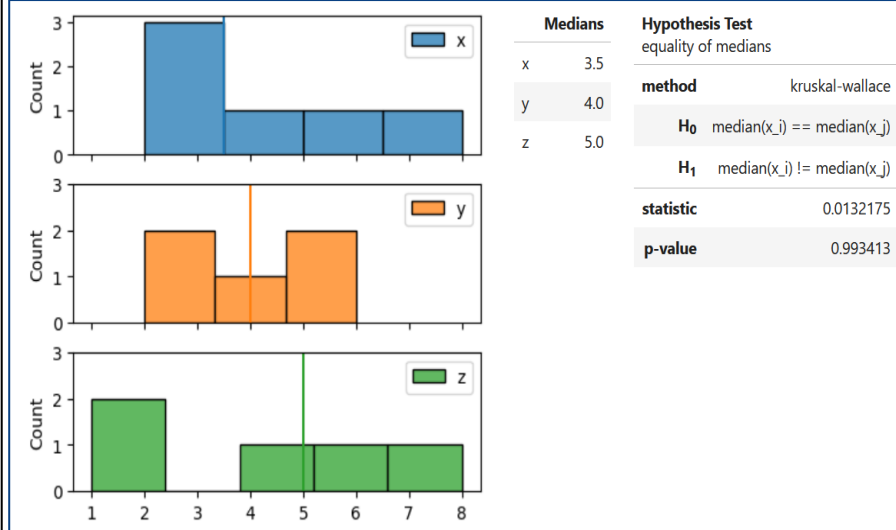
Example 3: Kruskal-Wallis Test – Python

```
x = np.array([4, 2, 3, 6, 8, 2])
y = np.array([6, 3, 4, 5, 2])
z = np.array([5, 2, 6, 1, 8])
samples = [x, y, z]
names = ['x', 'y', 'z']
colors = ['C0', 'C1', 'C2']

with Figure(4, 4, len(samples), sharex=True) as (fig, axes):
    for sample, name, color, ax in zip(samples, names, colors, axes):
        sns.histplot(data=sample, binwidth=1.5, color=color, ax=ax)
        ax.legend(name)
        ax.set_yticks([0, 1, 2, 3])
        ax.axvline(np.median(sample), color=color)
    plot = mqr.nbtools.grab_figure(fig)

medians = f'''
| | Medians |
|---|---|
| x | {np.median(x)} |
| y | {np.median(y)} |
| z | {np.median(z)} |
'''

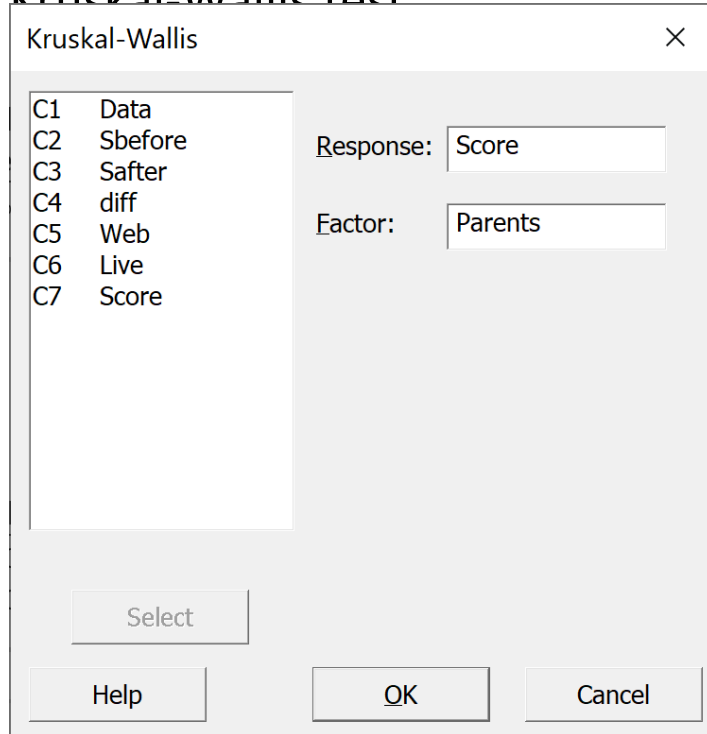
hypptest = mqr.inference.nonparametric.median.test_nsample(
    x, y, z,
    method='kruskal-wallace')
hstack(plot, medians, hypptest)
```



p is not less than α .
 We cannot reject H_0
 We cannot reject $\eta_1 = \eta_2 = \eta_3$.
 Education of parents doesn't matter.

Example 3: Kruskal-Wallis Test – Minitab

Run a Kruskal-Wallis test



The dialog box for the Kruskal-Wallis test in Minitab. It shows a list of variables on the left: C1 Data, C2 Sbefore, C3 Safter, C4 diff, C5 Web, C6 Live, and C7 Score. The 'Response' field is set to 'Score' and the 'Factor' field is set to 'Parents'. There are buttons for 'Select', 'Help', 'OK', and 'Cancel'.

Descriptive Statistics

Parents	N	Median	Mean Rank	Z-Value
C	5	4.0	8.4	-0.06
GS	5	5.0	8.7	0.11
HS	6	3.5	8.4	-0.05
Overall	16		8.5	

Test

Null hypothesis	H_0 : All medians are equal		
Alternative hypothesis	H_1 : At least one median is different		
Method	DF	H-Value	P-Value
Not adjusted for ties	2	0.01	0.994
Adjusted for ties	2	0.01	0.993

p is not less than α .

We cannot reject H_0

We cannot reject $\eta_1 = \eta_2 = \eta_3$.

Education of parents doesn't matter.

Non-Parametric Hypothesis Tests

Nonparametric Test	Test For	Parametric equivalent	# of Factors	# of Levels
Run test	Randomness	Distribution Fit Tests (Normality, ...)	1	
1-Sample Sign	Central Location	1 sample t Paired t	1	1
1-Sample Wilcoxon			1	2
Man-Whitney	Equality of Two Population Locations	2 sample t	1	2
Kruskal-Wallis	Equality of Multiple Population Locations	ANOVA	1	2+
Moods Median			1	2+
Friedman	Randomized Block Experiment	2 way ANOVA	2	2+
Levene's	Dispersion	Chi-2 test	1	1
		F-test Bartlett's test	1	2+



Summary

Statistical tests are those with random outcomes.

There are two types of test errors with alpha and beta risks.

The t-test is used to make decisions based on the population mean.

Other statistical tests are used to make decisions about other population parameters.

Necessary sample sizes for a test can be determined beforehand using calculators

Non-parametric statistical tests exist for data that is not normal or distribution fit

Next Topic: Quality Management in Manufacturing